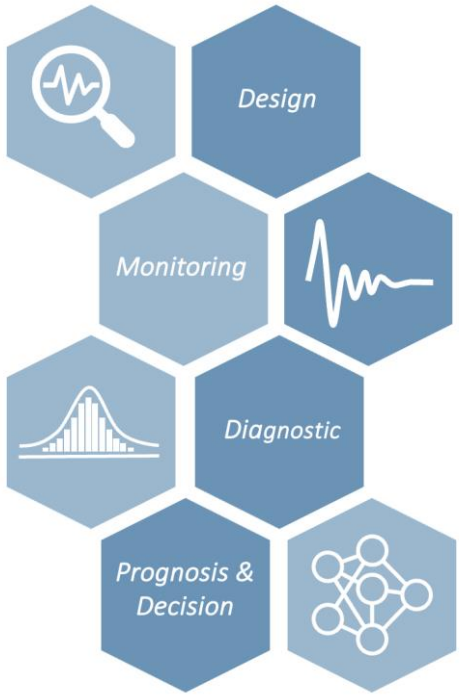
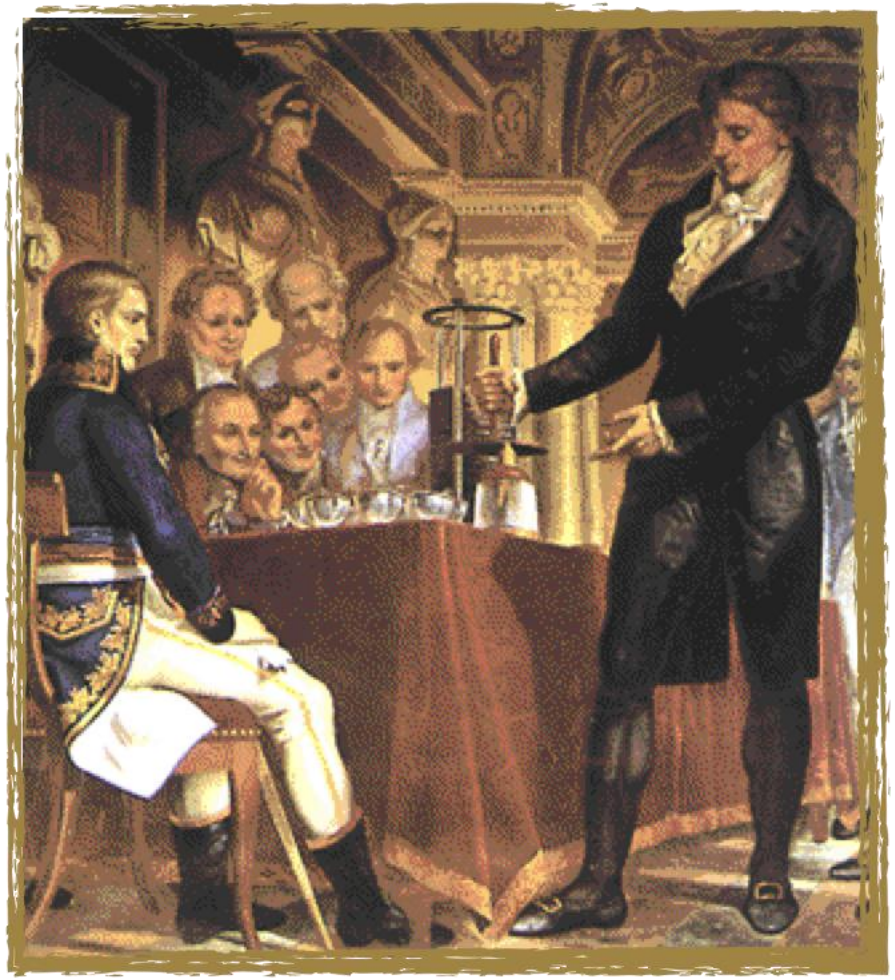
Ingénierie augmentée par les données, l'IA et les jumeaux numériques



Francisco (Paco) CHINESTA
Francisco.Chinesta@ensam.eu

DISCLAIMER



Alessandro Volta

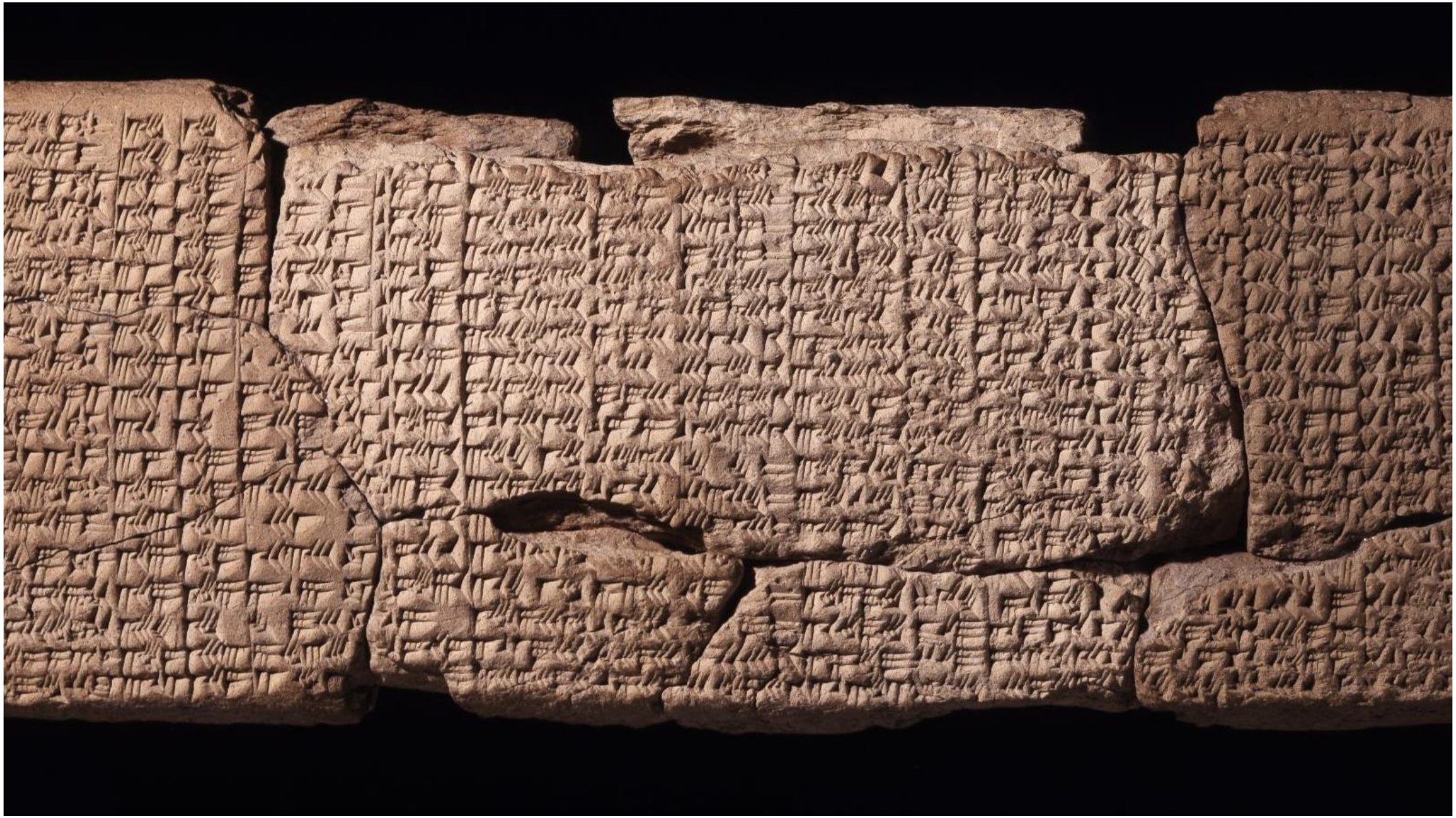
Académie de Sciences

Paris 1801

Babylonians: data acquisition



Data storage



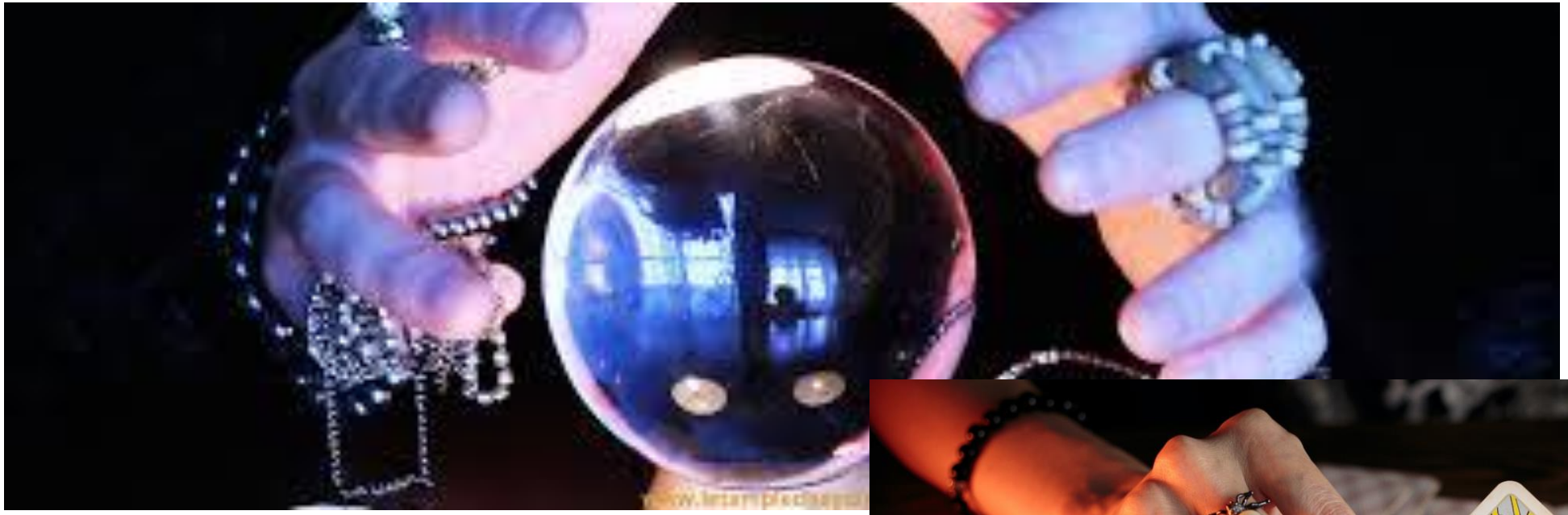
Rulers … for everything …

musée des arts et métiers



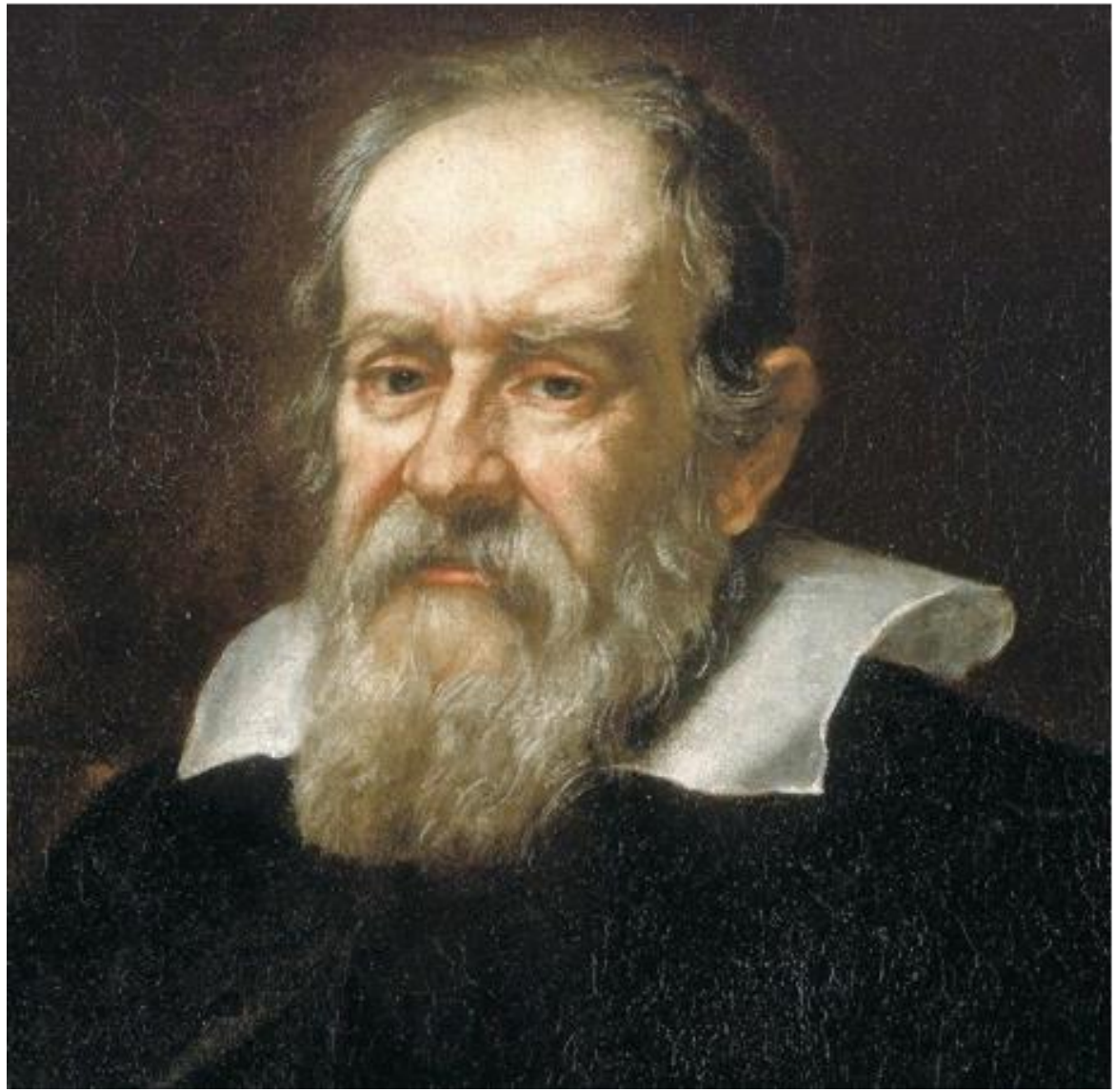


Then, all you need is predicting …





Learning for predicting: Galileo









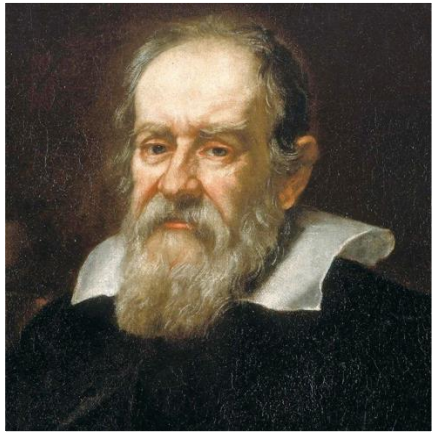
$$\begin{cases} t=1,\ \ v=10,\ \Delta x=5 \\ t=2,\ v=20,\ \Delta x=15 \\ t=3,\ v=30,\ \Delta x=25 \\ t=4,\ v=40,\ \Delta x=35 \\ \qquad\qquad \cdots \end{cases}$$

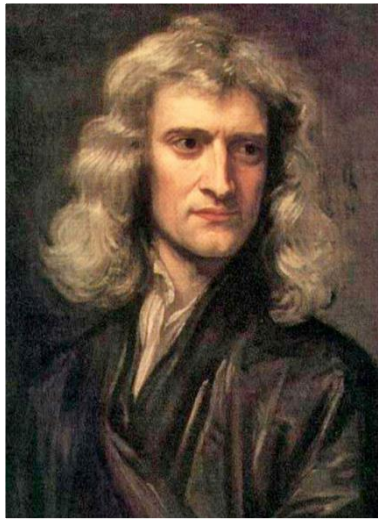
$$5/15 = 1/3$$

$$15/25 = 3/5$$

$$25/35 = 5/7$$

The two faces of a unique reality





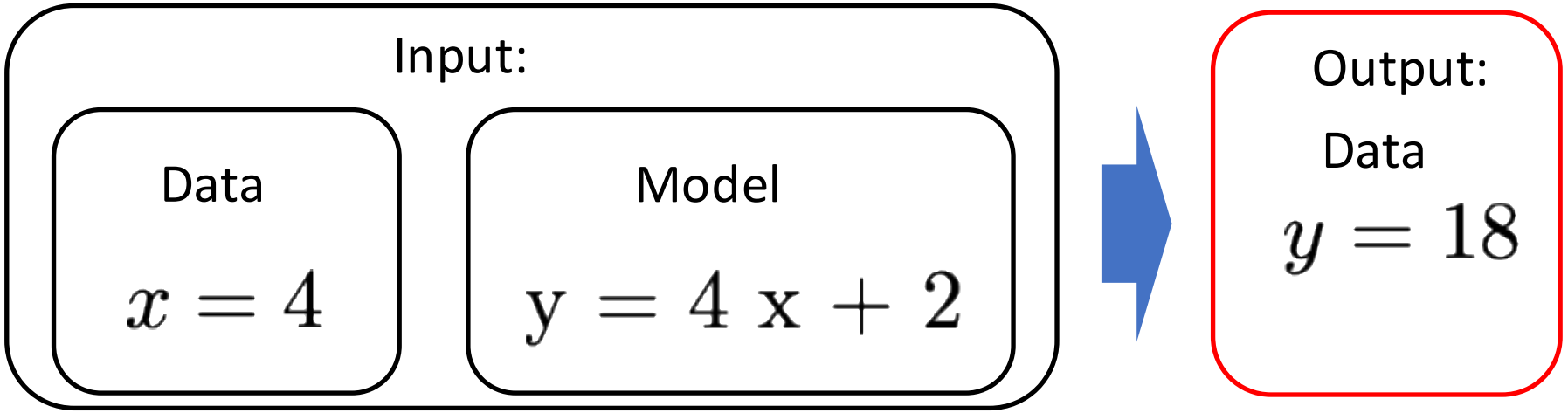
$$5/15 = 1/3$$
$$15/25 = 3/5$$
$$25/35 = 5/7$$



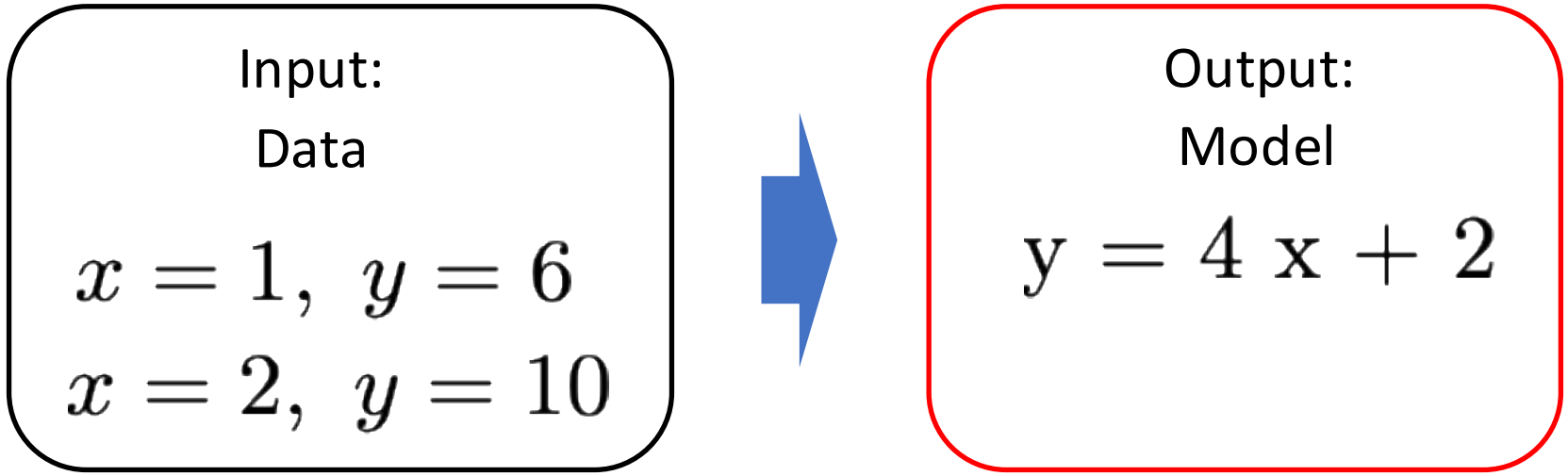
$$F_g = m\frac{d^2x}{dt^2}$$

Simulation versus Learning

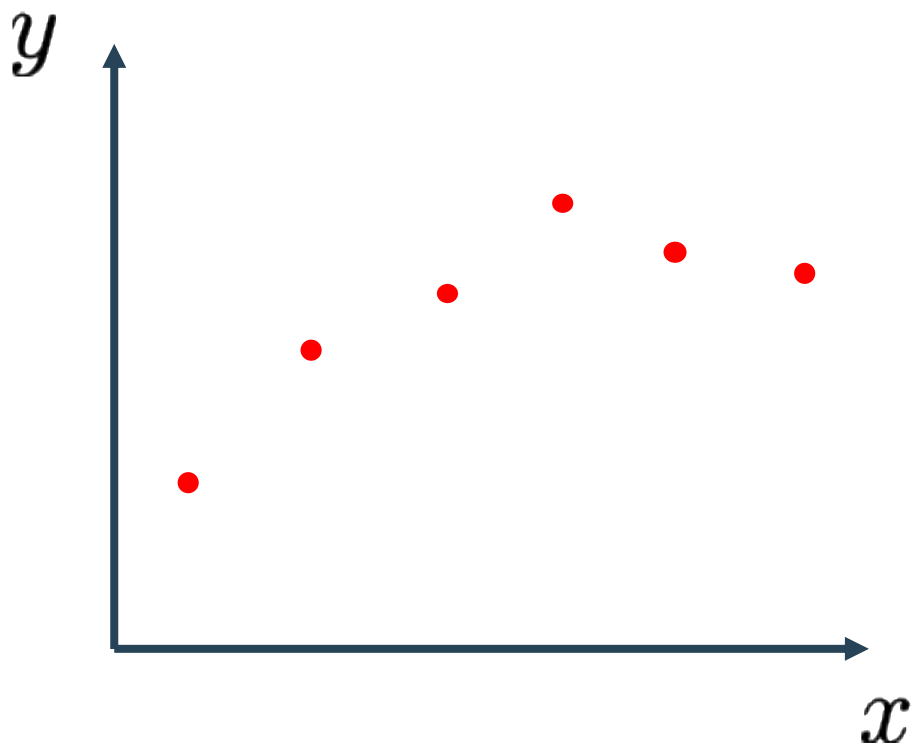
Simulation:

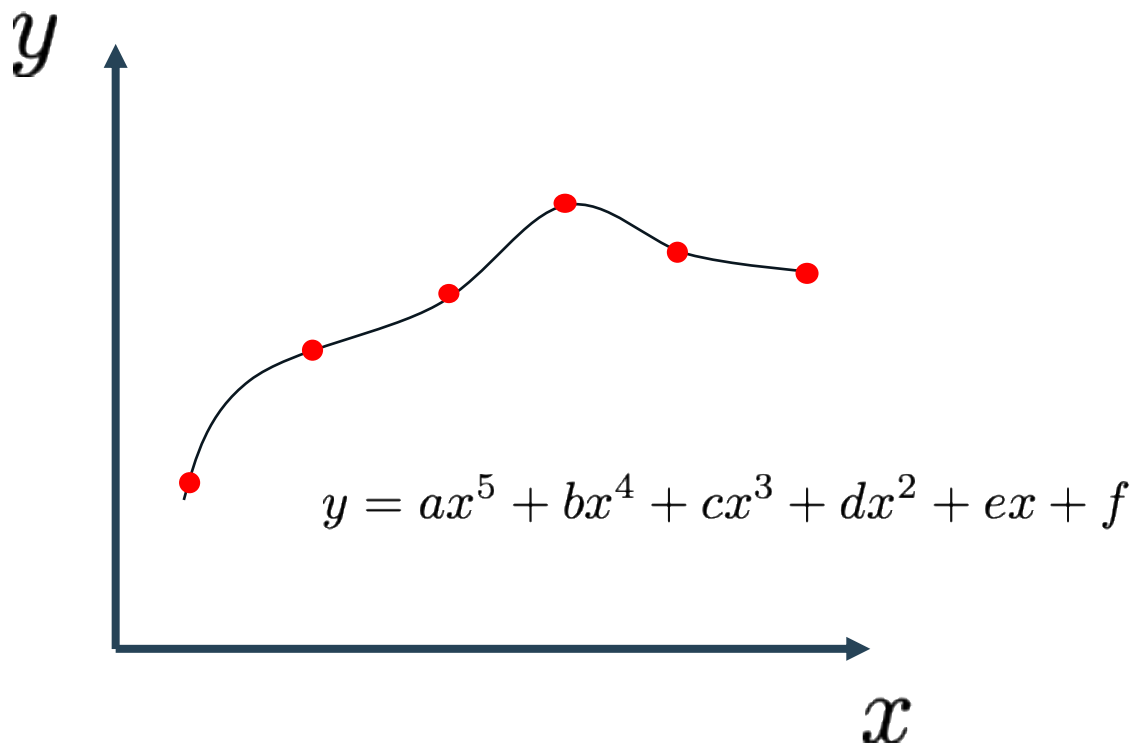


Learning:

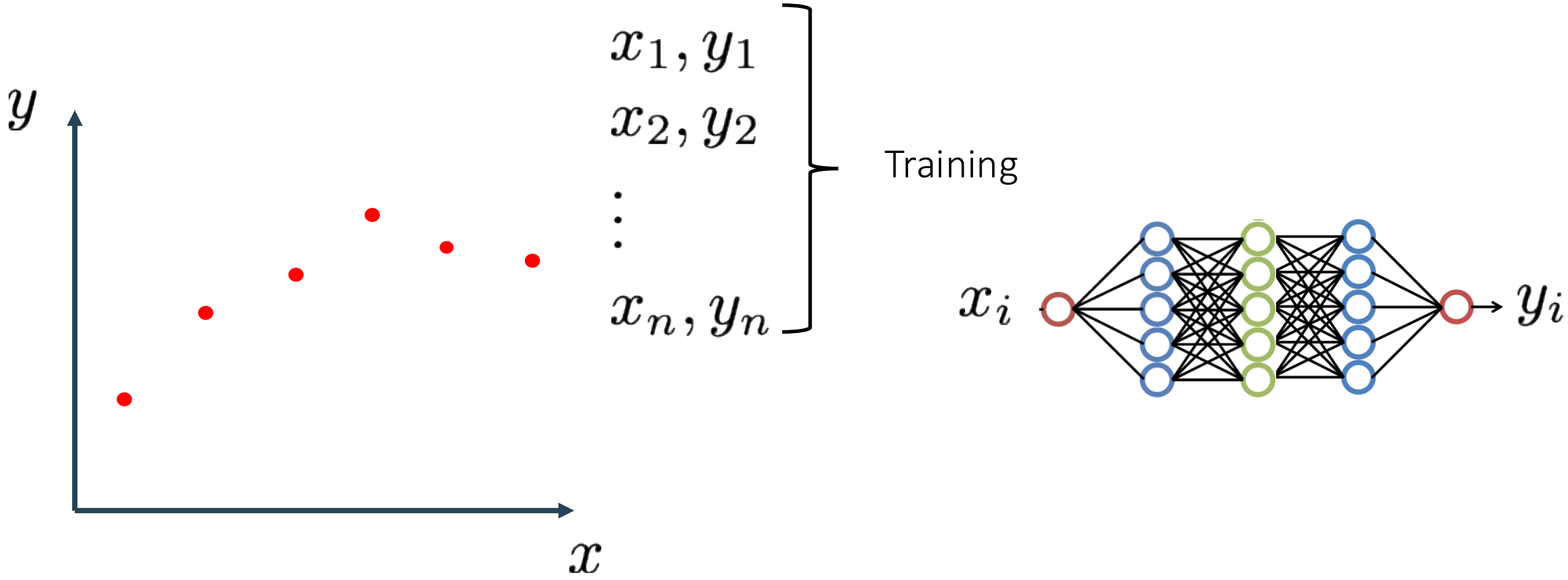


Learning: polynomial regression

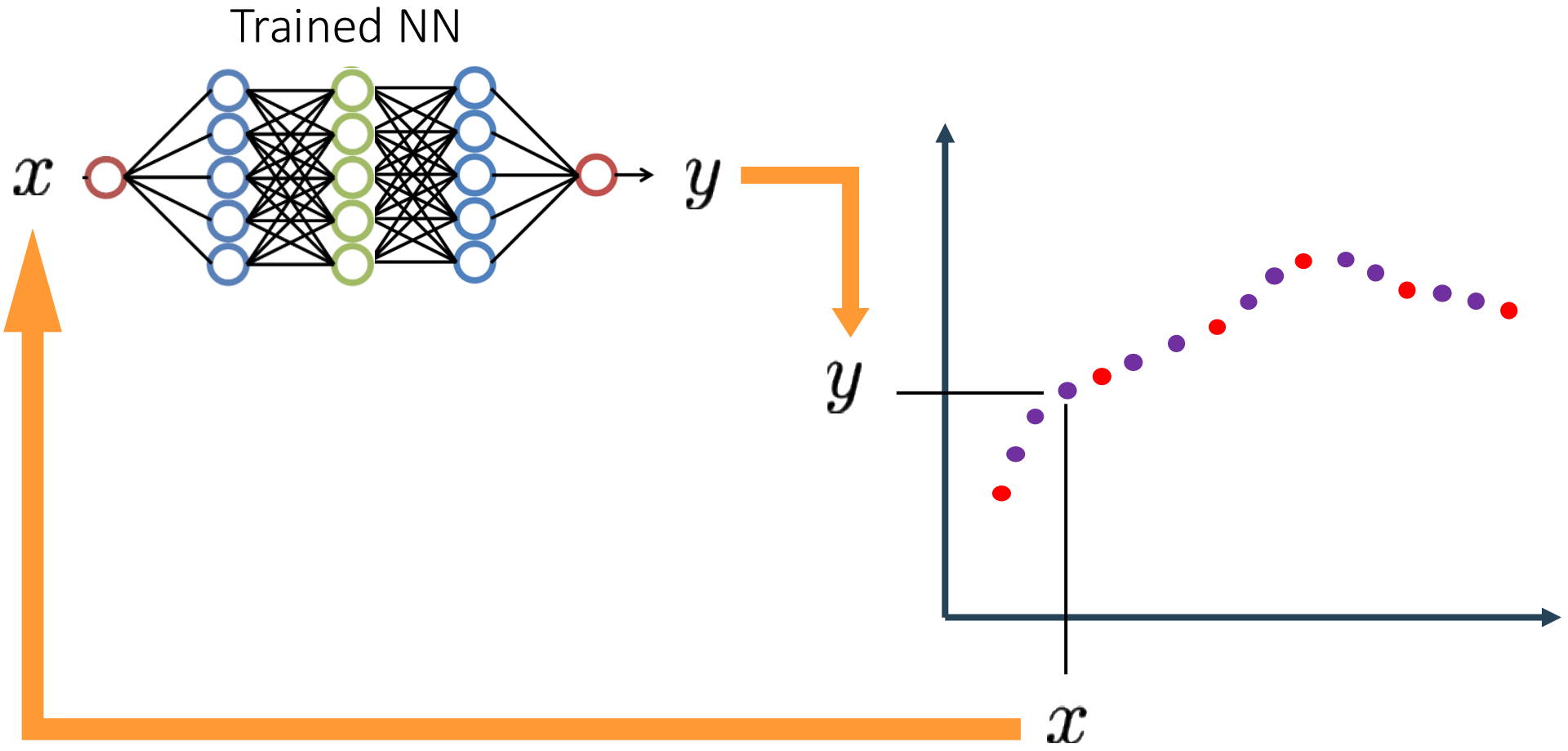




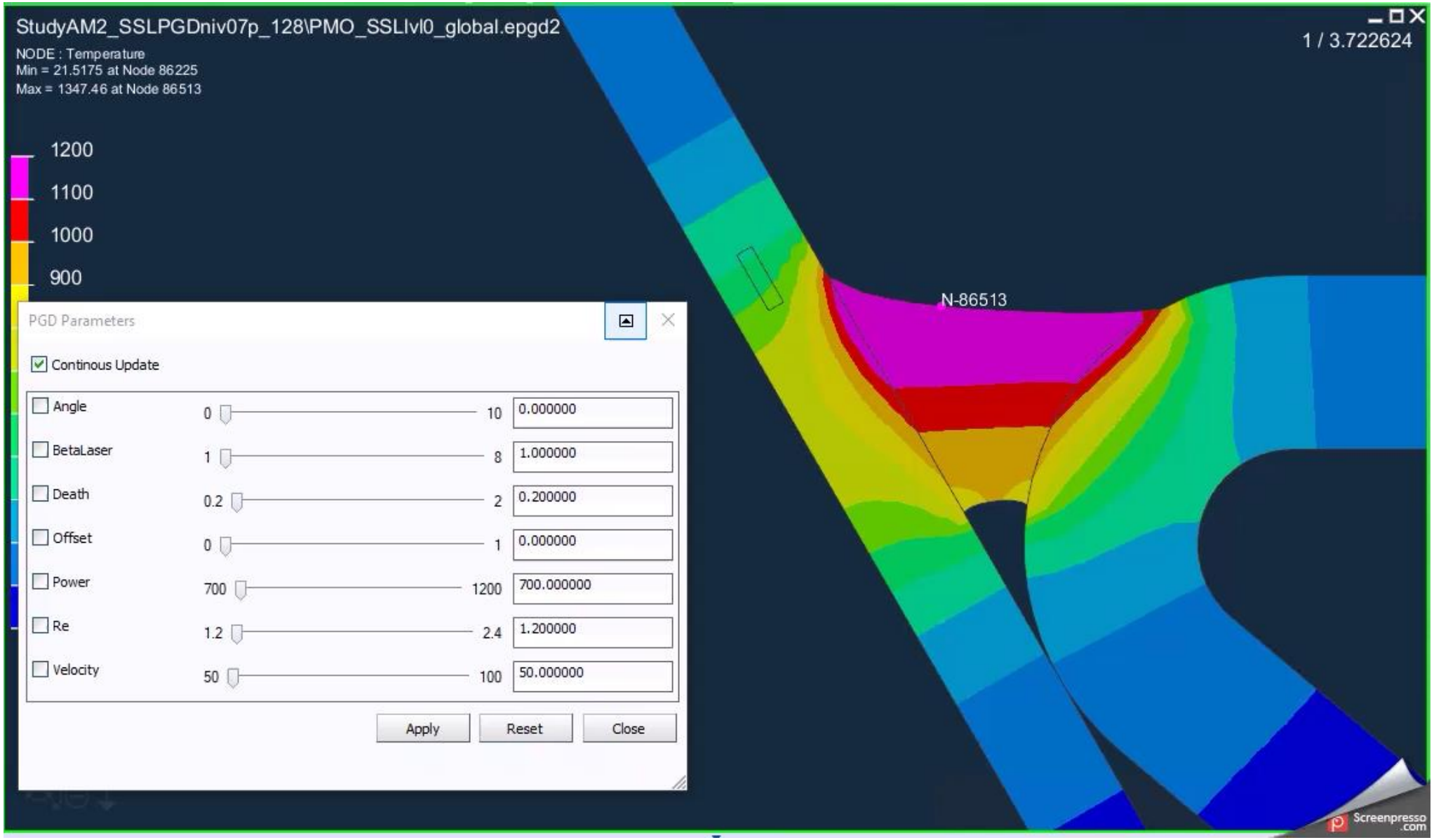
Learning: Neural Network -NN



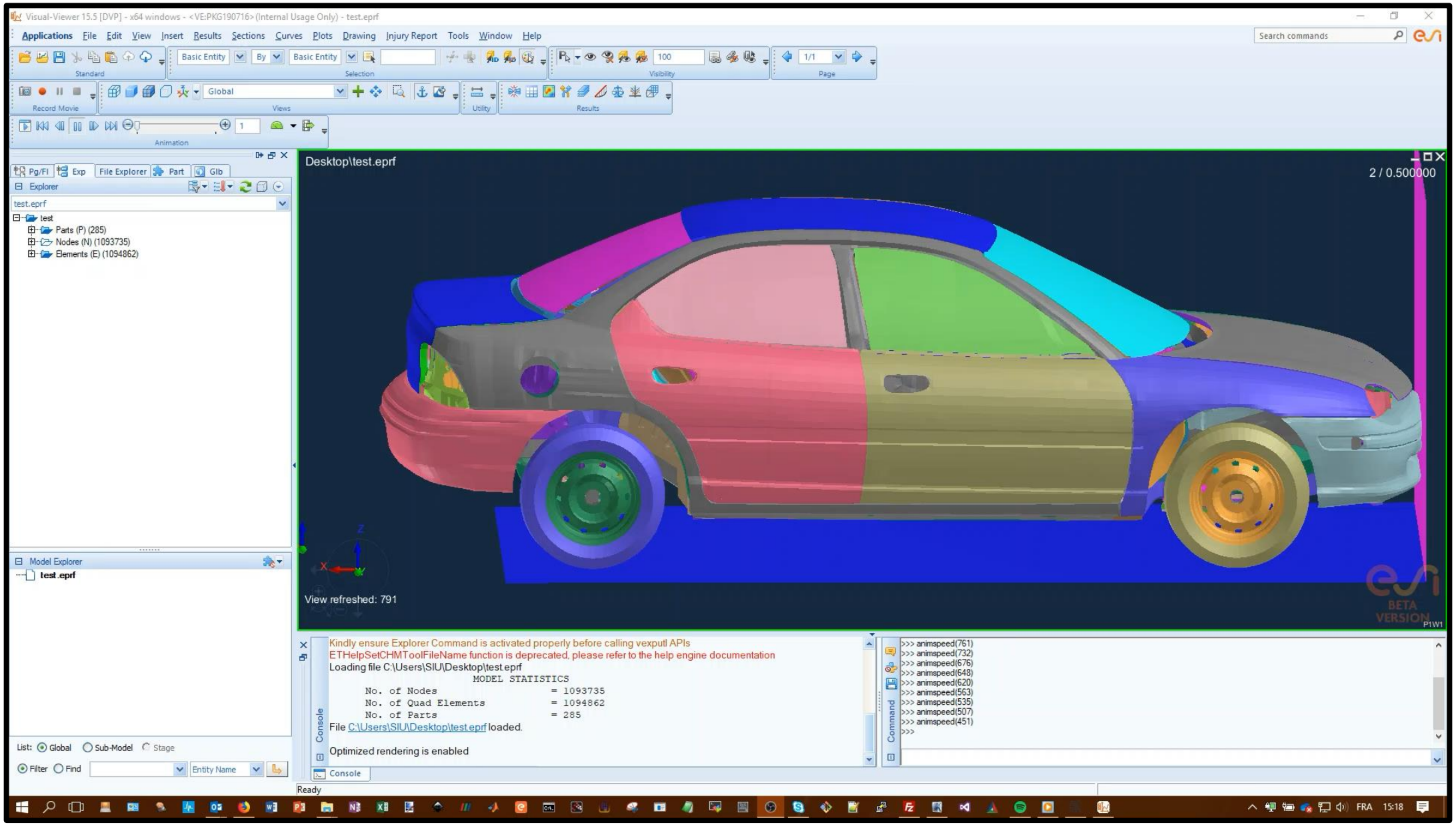
NN Inference



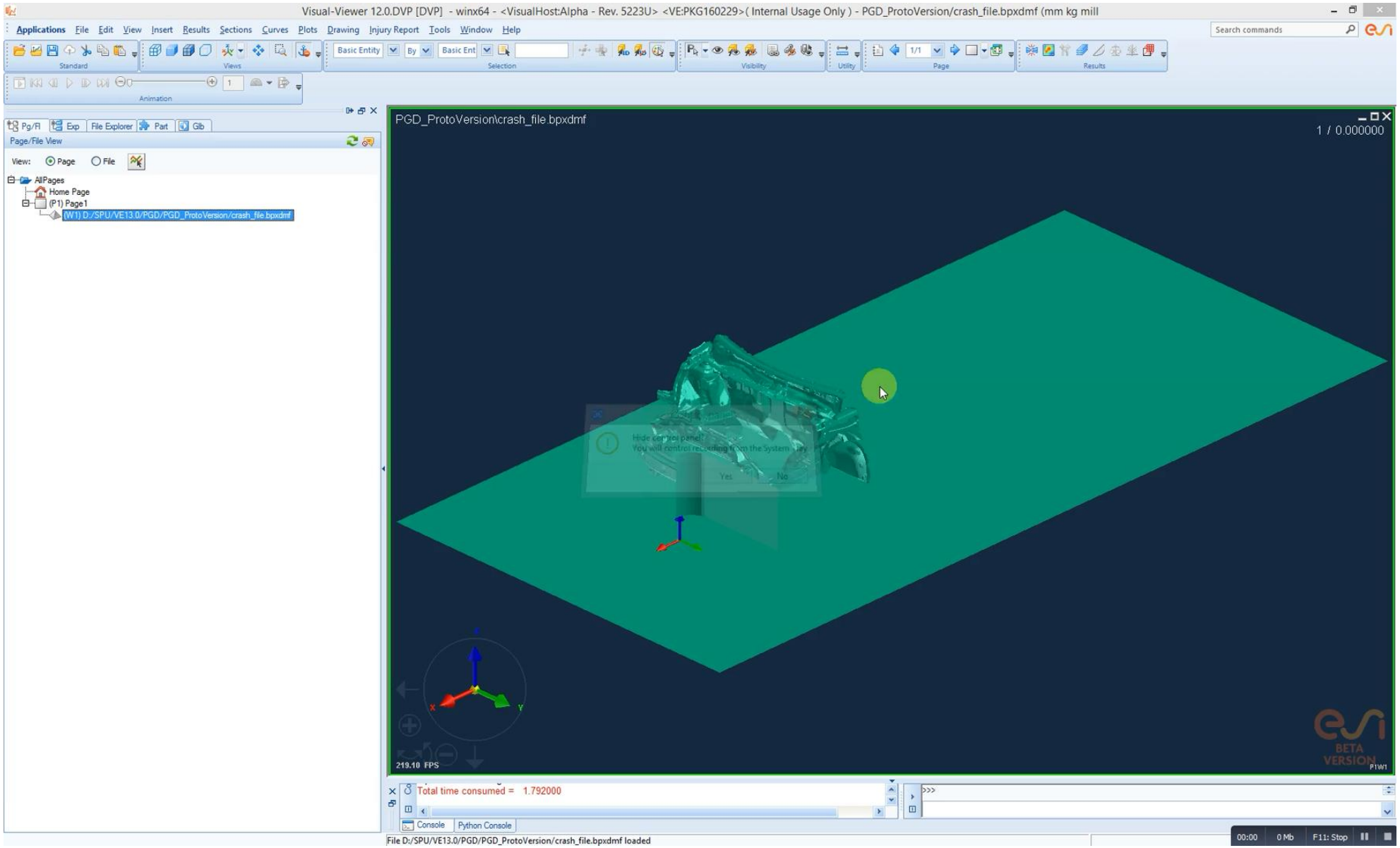
Parametric real-time physics



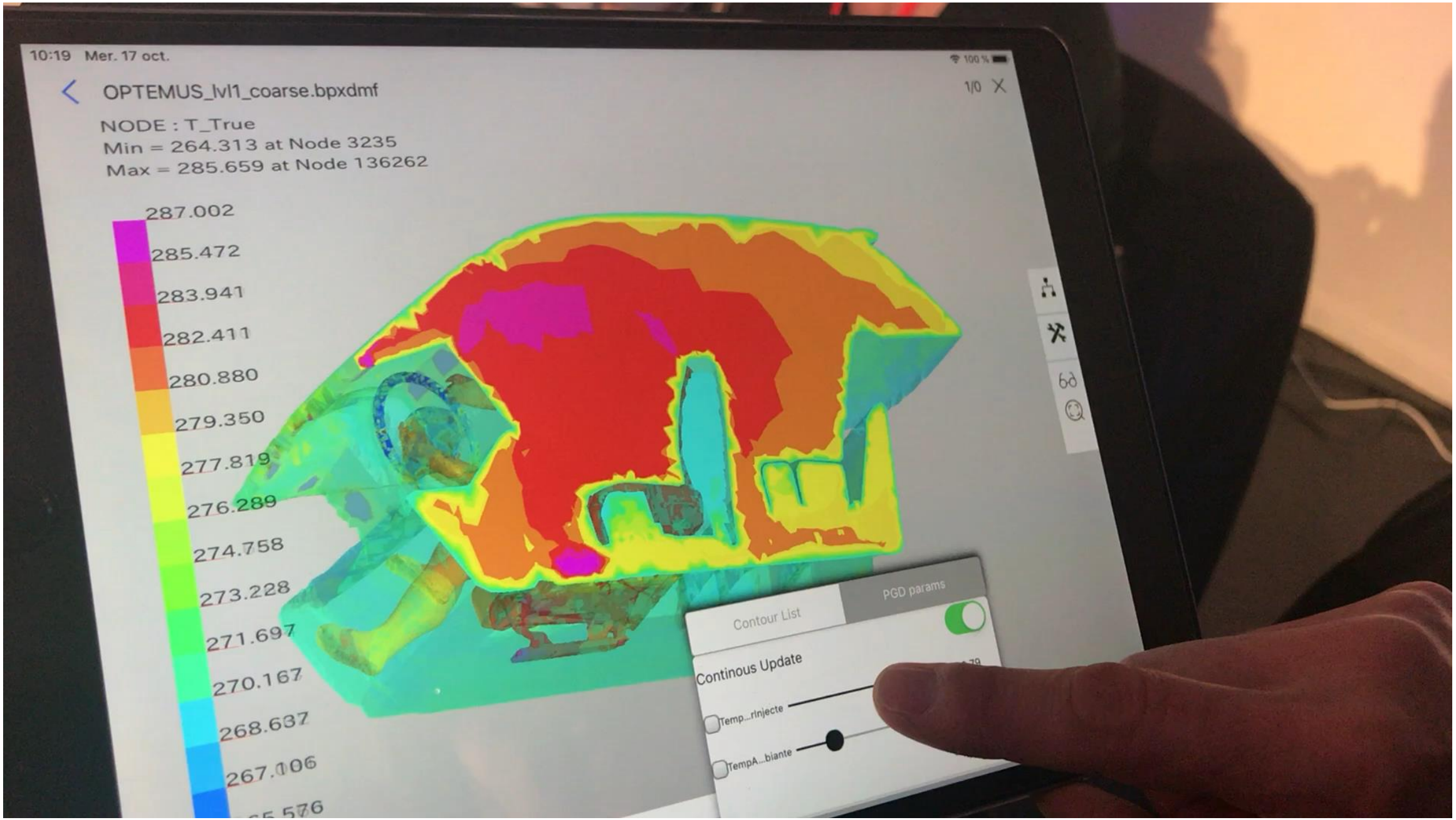
Parametric real-time physics



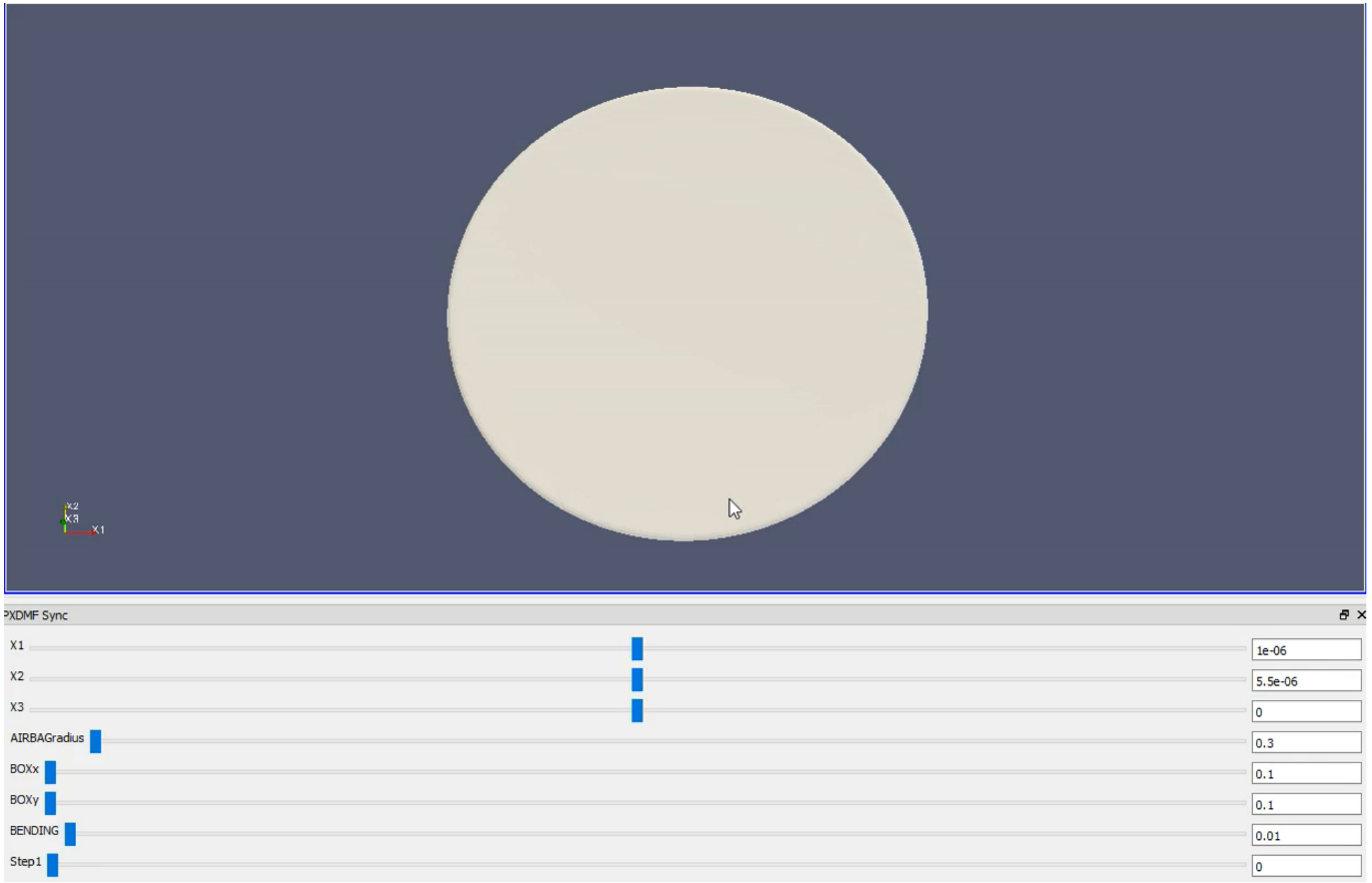
Parametric real-time physics



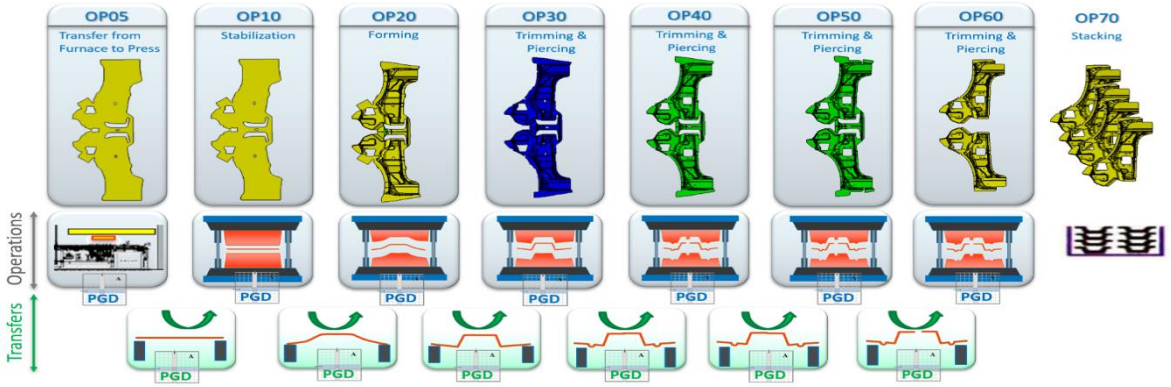
Parametric real-time physics

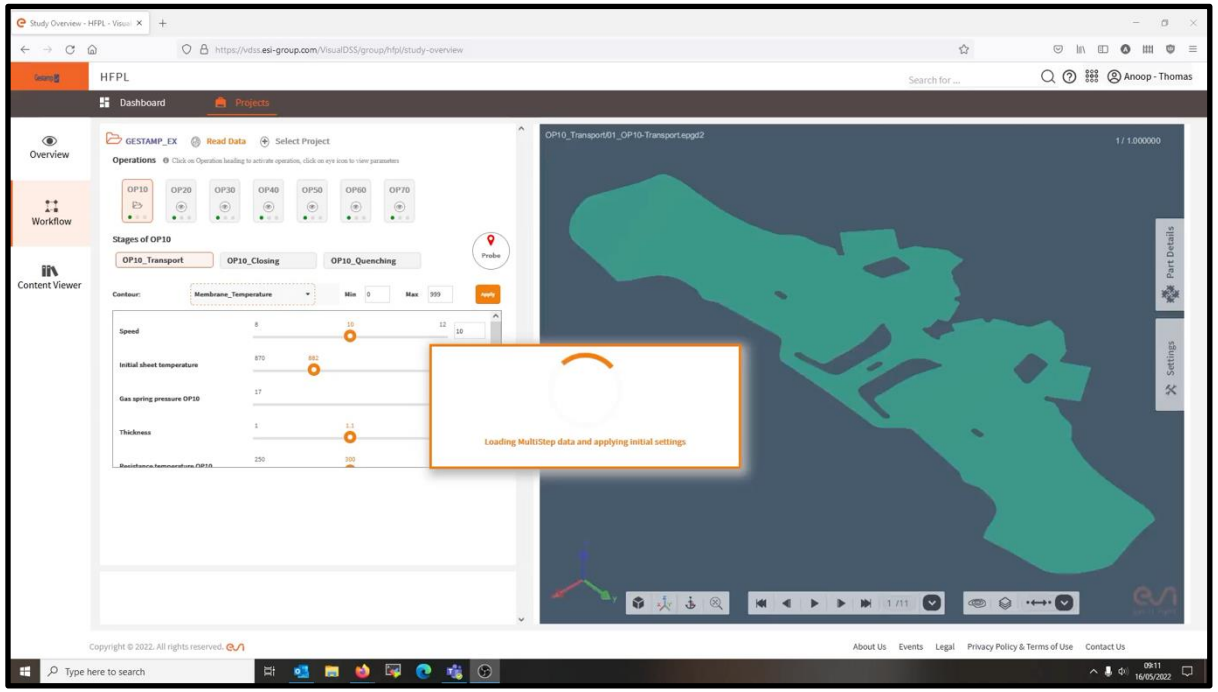


Parametric real-time physics



Parametric real-time physics





- COMPLEXITY -

10 steps & 4 parameters/step

100 values/parameter =

$= 10^{2x4x10} = 10^{80}$ combinations

10^{80} number of elementary particles in the universe

10^{80} s if one run takes 1 second

10^{21} s the universe age !

few hundred of runs (instead of 10^{80})

Parametric real-time physics



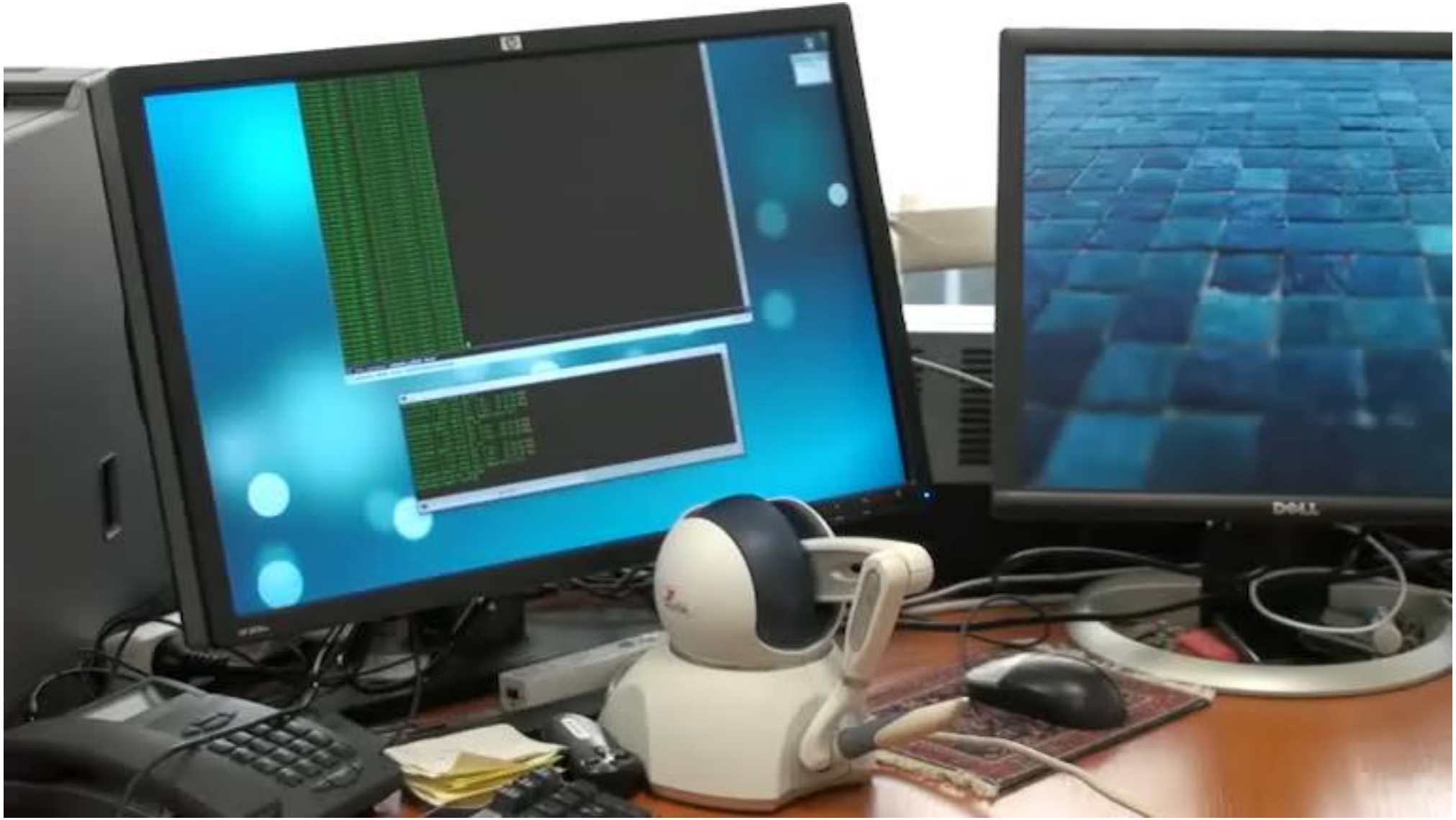
Physics-aware interaction between virtual and physical objects in Mixed Reality

A. Badías, D. González, I. Alfaro, F. Chinesta, E. Cueto

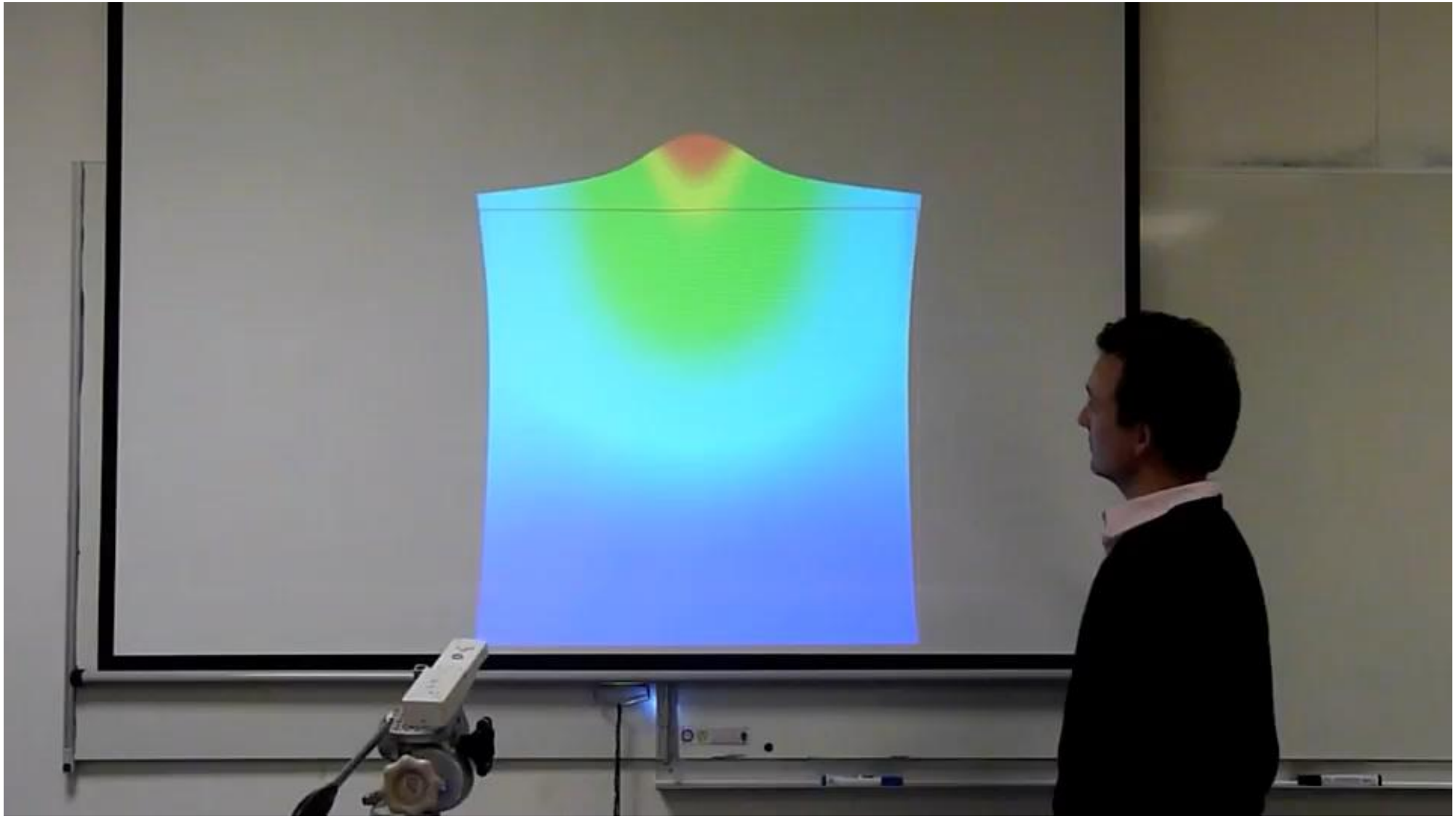
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Real-time interaction



Real-time interaction



Haptic rendering by thermodynamics-informed Graph Neural Networks

Q. Hernández, P. Martins, D. González, I. Alfaro, F. Chinesta, E. Cueto

ESI Group Chair, Universidad de Zaragoza



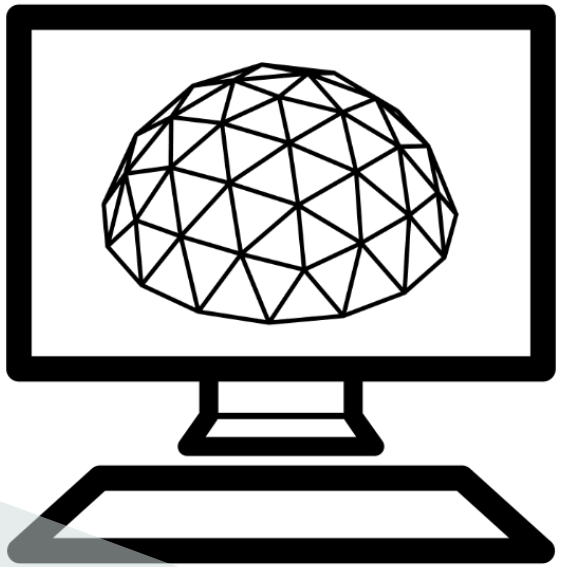


Universidad Zaragoza

Digital twin

THE THREE LEVELS OF DIGITALIZATION



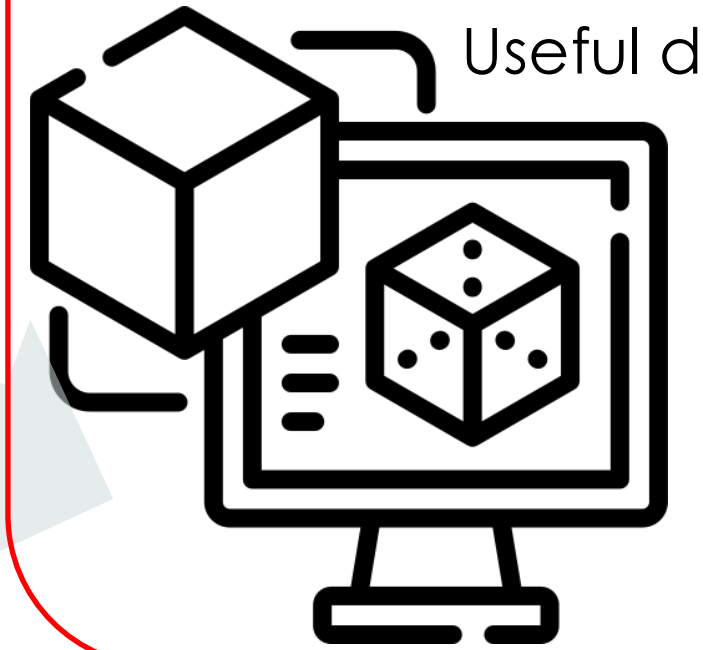


In design

In operation

Physical asset (or process)

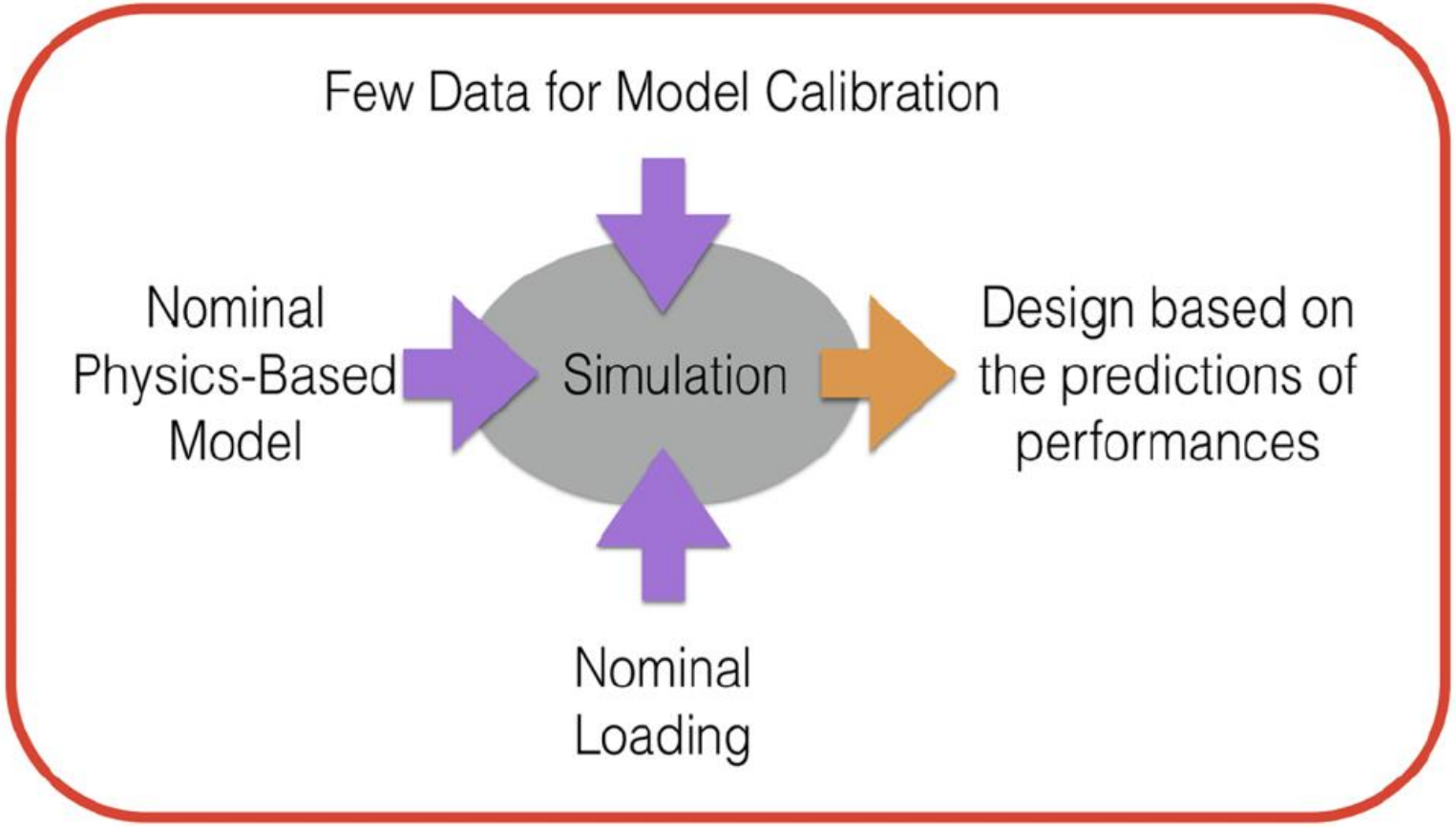
Useful data



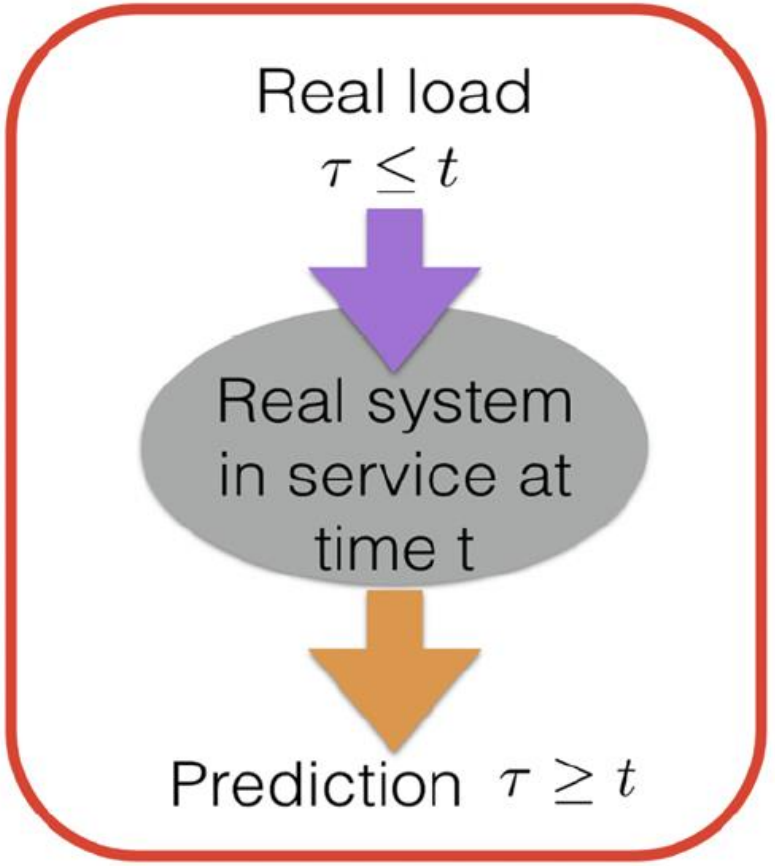
As built or
as practiced
Digital replica

THE TWO ENGINEERING: Design versus Operation

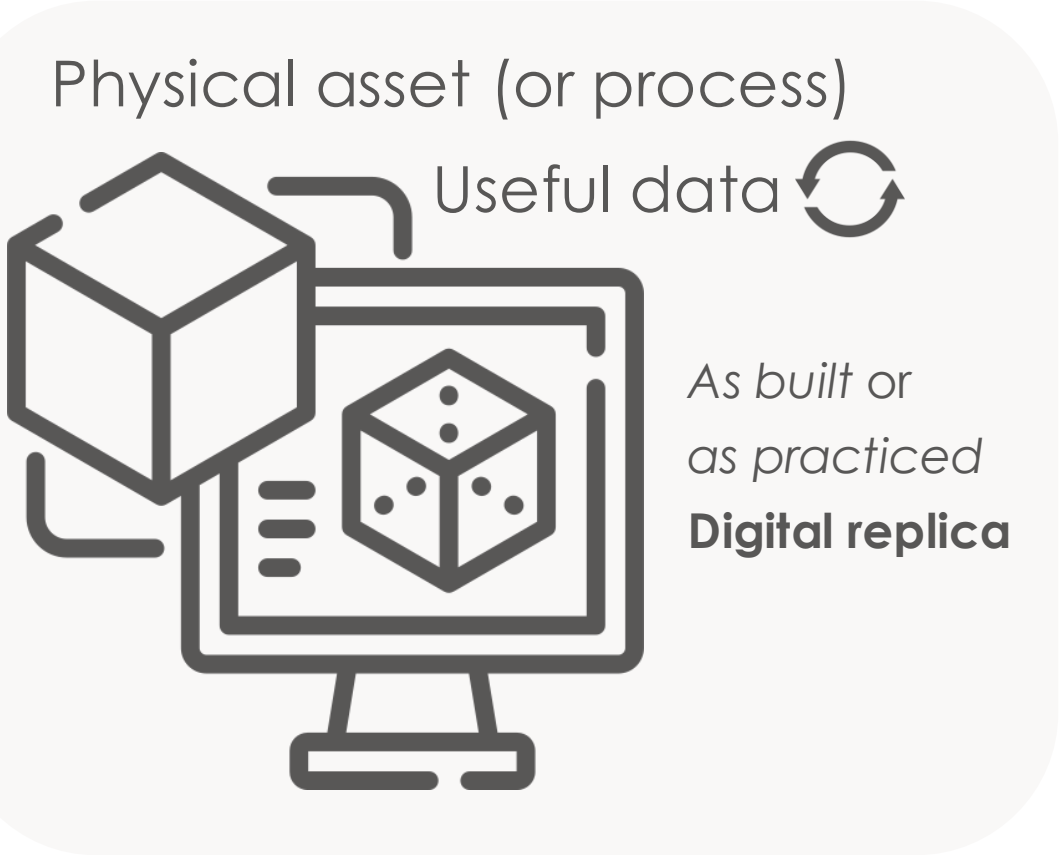
Design



Operation



DIGITAL TWIN: Anatomy and function

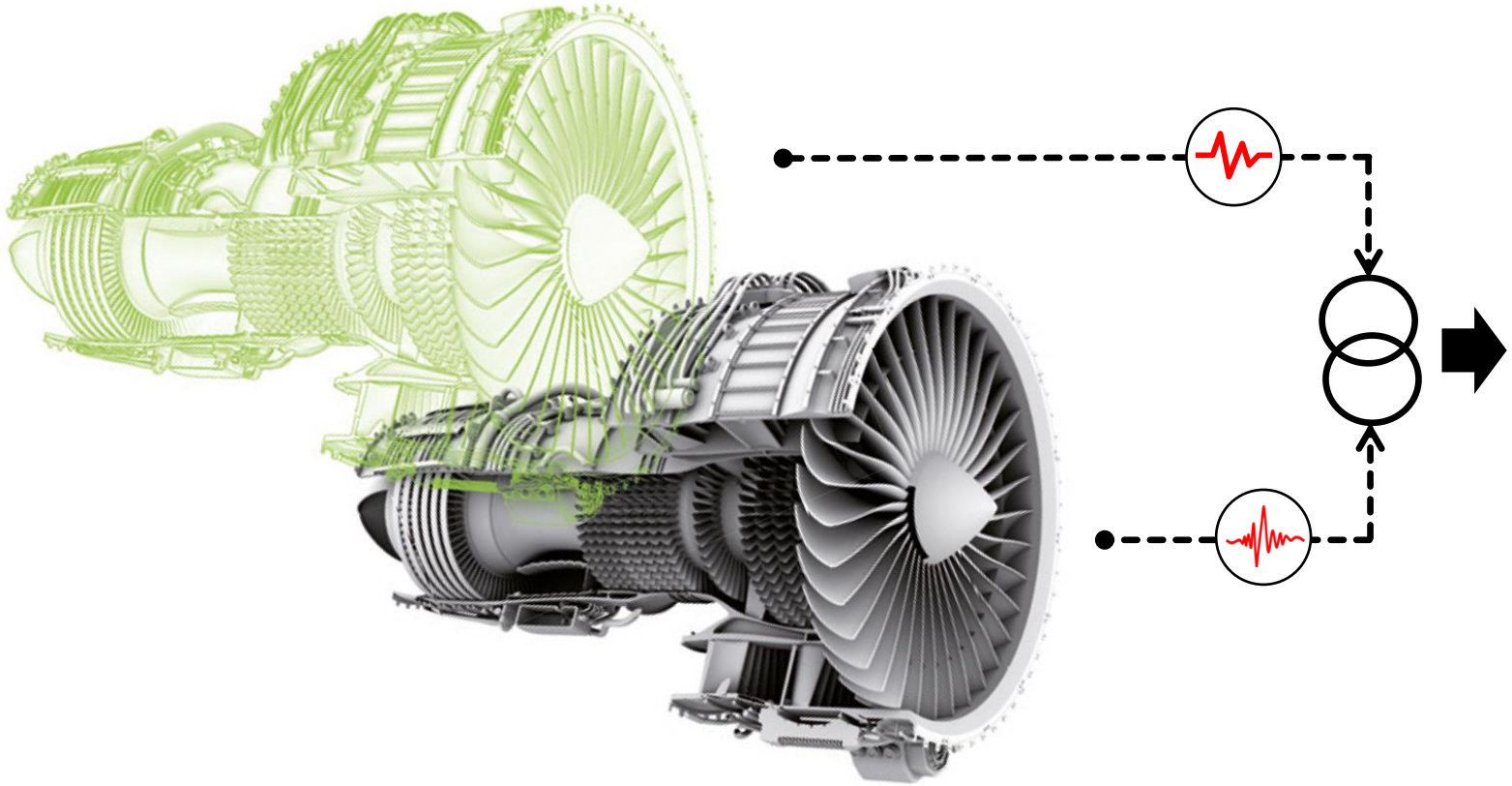


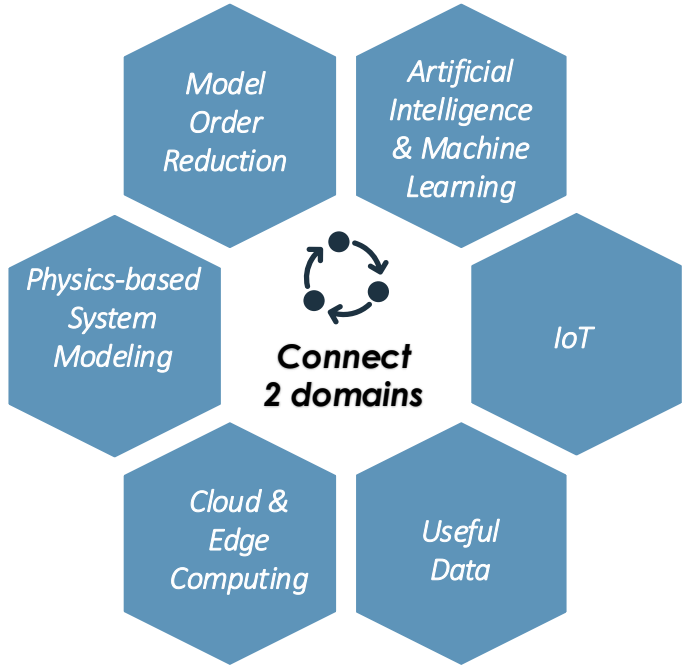
Based on:

- the best available multi-physics, multiscale & probabilistic computational models
- sensor data

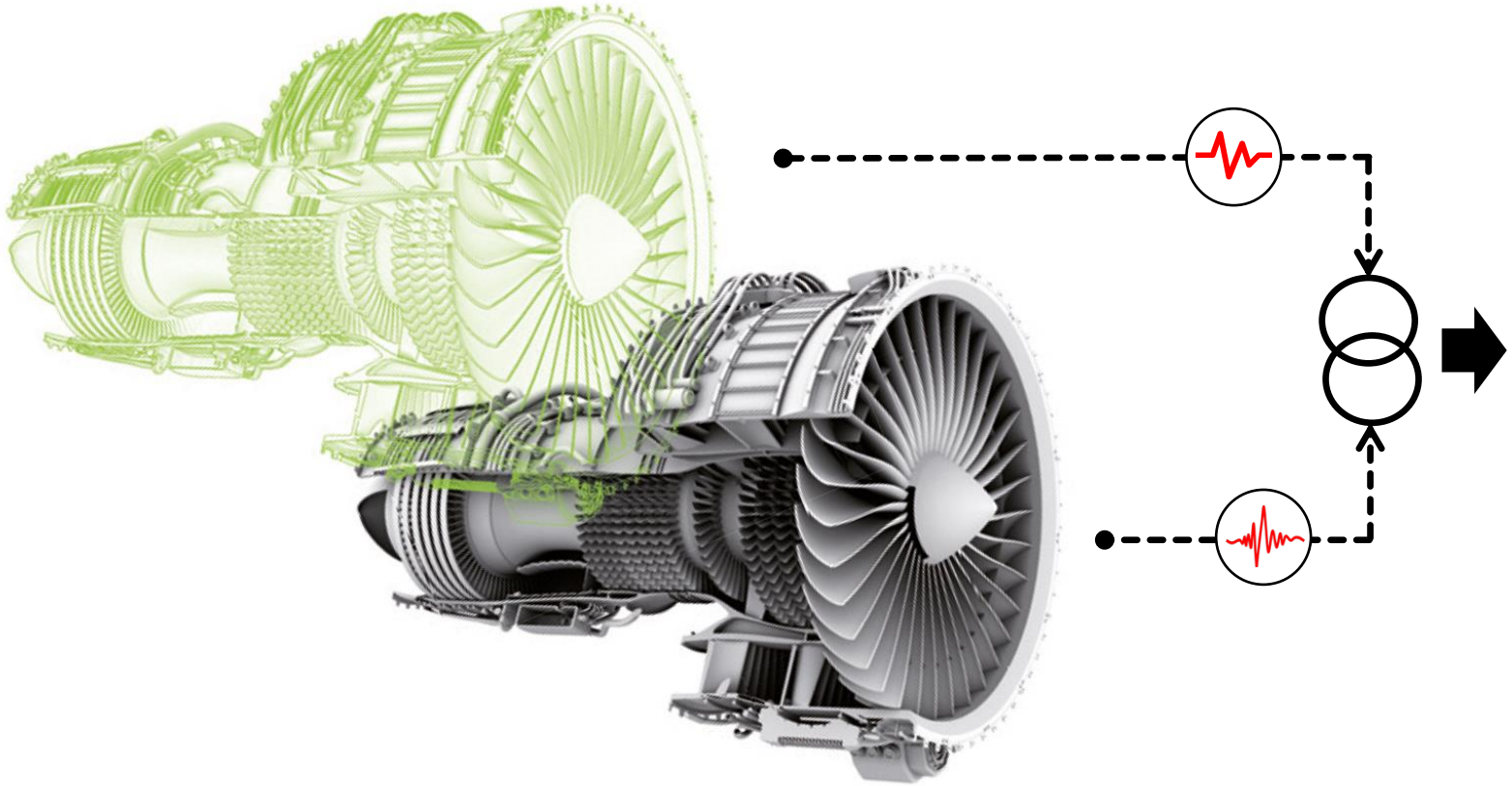
To mirror & predict the functioning and performances over the life cycle of the associated physical asset.

DIGITAL TWIN : Involved scientific domains and technologies





DIGITAL TWIN: Insights



Actionable Insights :

- Testing & Validation
- Optimization & Control
- Monitoring
- Diagnosis
- Prognosis
- Predictive Maintenance

THE LIMITS OF THE EXISTING MODELLING & SIMULATION PARADIGMS

Physics

Model

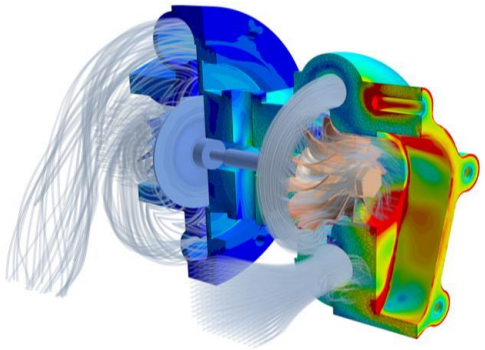
Data

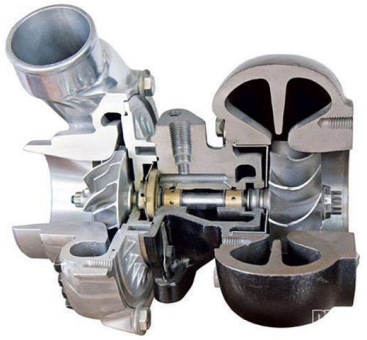


The usual simulation-based paradigm fails to perform diagnosis, prognosis and decision making when addressing complex systems of systems because of

- **Physics-based:** the lack of fidelity of state-of-the-art models, and the lack of efficiency related to their solution procedures.
- **Data-driven:** the availability of data, its quality, as well as the limitations related to the extrapolation or the ability to explain the predictions offered by the trained models.

The twin continuum





21st century engineering:

- From design to operation

- From component to system of systems

The Twin Continuum

VIRTUAL

- Mostly physics
- U=U(F)
- Software
- Processing time
- Resources
- Complex systems
- Variability
- Uncertainty

INFORMED

u(x) AI-based such that L(u)=G

AUGMENTED - HYBRID

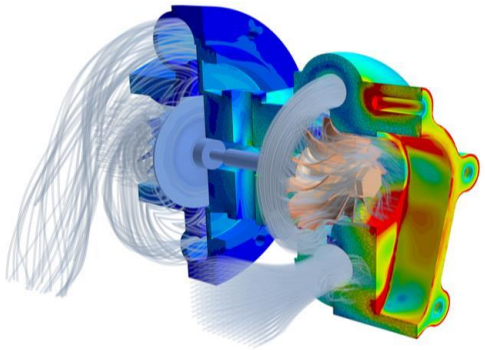
$u(x)=u^P(x)+u^D(x)$

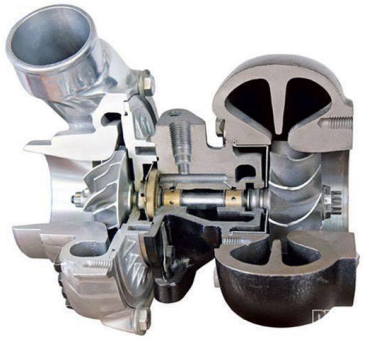
- Less data
- Faster
- Explainable

DIGITAL

- Unknown or poor model
- Ui,Fi, i=1,2, …
- Machine learning: U(F)
- Data cost
- Which data, where & when?
- Extrapolation
- Explanation

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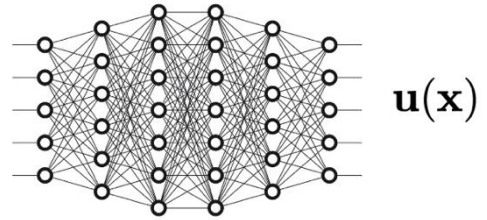
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Physics Informed NN - PINN

First order model $\mathcal{L}(u) = f(x)$

Unknown field $u(x)$

NN-based approximation $u = \mathcal{NN}(x)$ 



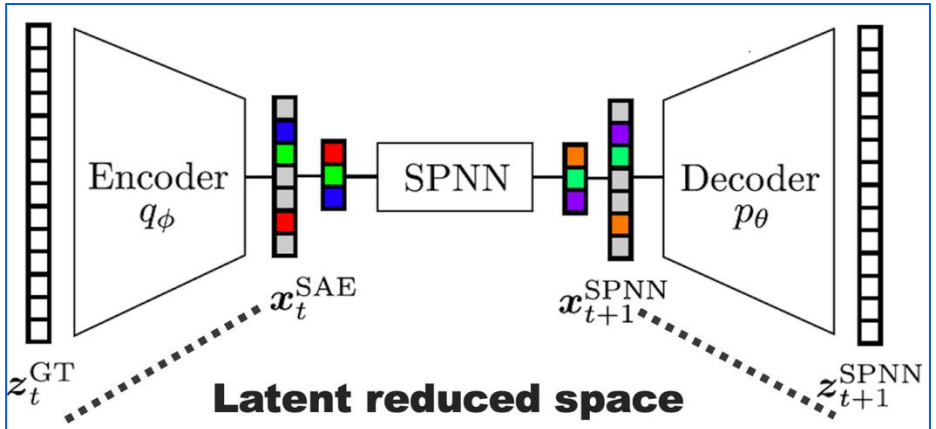
Collocation points where model is enforced $\{x_1^c, x_2^c, ..., x_C^c\}$

Observation points $\{x_1^o, x_2^o, ..., x_O^o\}$

where measures apply $\{u_1^o, u_2^o, ..., u_O^o\}$

Training loss: $\texttt{Loss} = \sum_{i=1}^{C} \{\mathcal{L}(u)|_{x_i} - f(x_i)\}^2 + \sum_{j=1}^{O} \{u(x_j) - u_j^o\}^2 + \text{BC}$

Thermodynamics Informed NN - TINN



GENERIC

$$\dot{z}_t = \boldsymbol{L}(z_t)\nabla E(z_t) + \boldsymbol{M}\nabla S(z_t), \quad z(0) = z_0$$

Poisson matrix: reversibility

Friction matrix: irreversibility

with $\boldsymbol{L}(z)\cdot\nabla S(z) = \mathbf{0}$, $\boldsymbol{M}(z)\cdot\nabla E(z) = \mathbf{0}$.

$$\frac{z_{n+1}-z_n}{\Delta t} = \mathsf{L}(z_{n+1}, z_n)\mathsf{DE}(z_{n+1}, z_n) + \mathsf{M}(z_{n+1}, z_n)\mathsf{DS}(z_{n+1}, z_n)$$

$$\boldsymbol{\mu} = \{\mathsf{L}, \mathsf{M}, \mathsf{DE}, \mathsf{DS}\} = \arg\min_{\boldsymbol{\mu}^*} ||\boldsymbol{z}(\boldsymbol{\mu}) - \boldsymbol{z}^{\text{meas}}||$$

with $\mathsf{DE} = \boldsymbol{A}\boldsymbol{z}$, $\mathsf{DS} = \boldsymbol{B}\boldsymbol{z}$

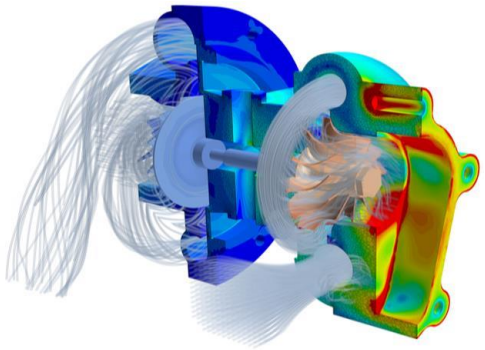
Physically sound, self-learning digital twins for sloshing fluids

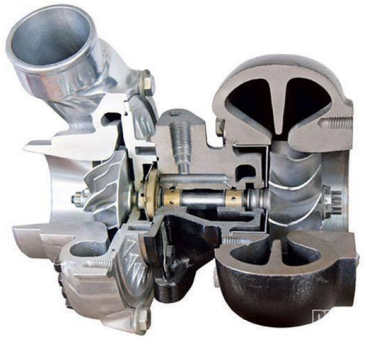
B. Moya, I. Alfaro, D. González, F. Chinesta, E. Cueto

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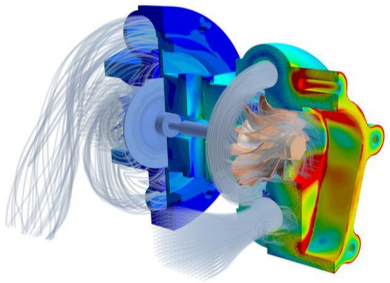
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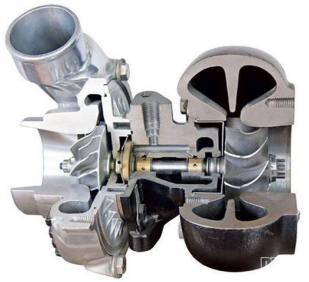
$u(x)=u^P(x)+u^D(x)$

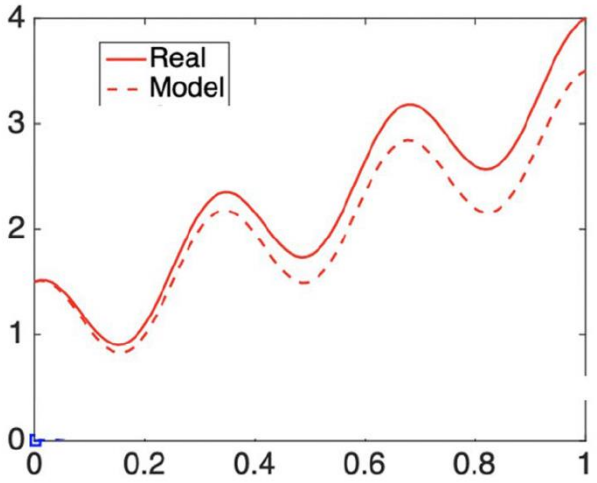
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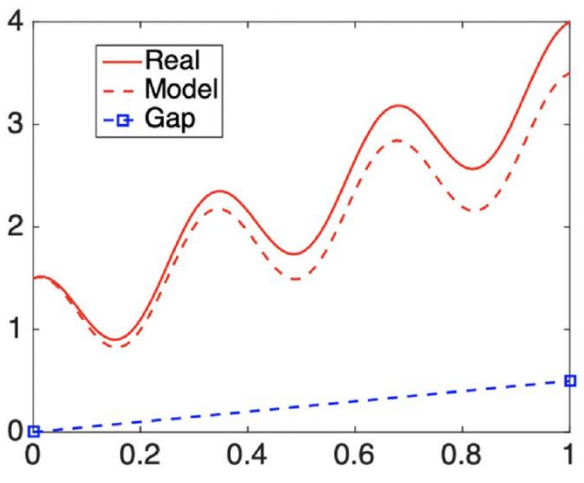
DIGITAL

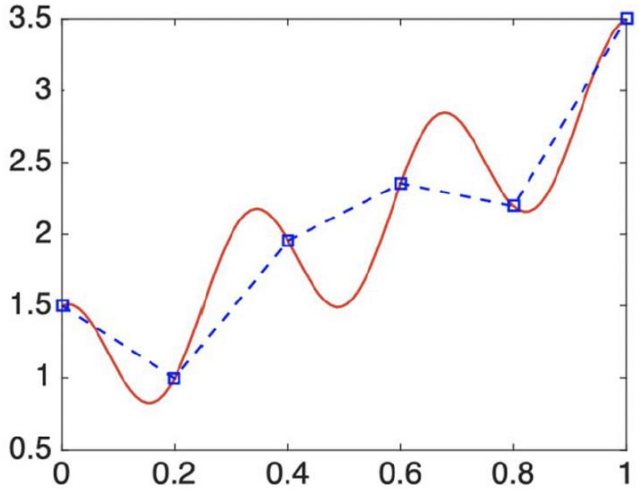
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- Machine learning: U(F)

- Data cost
- Which data, where & when?
- Extrapolation
- Explanation



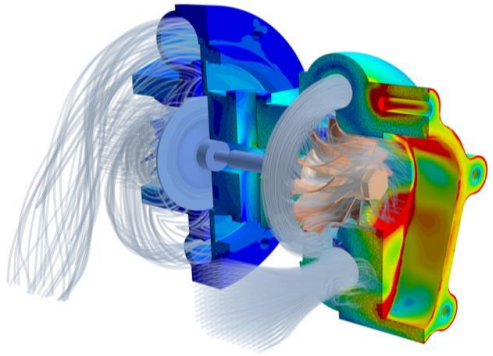


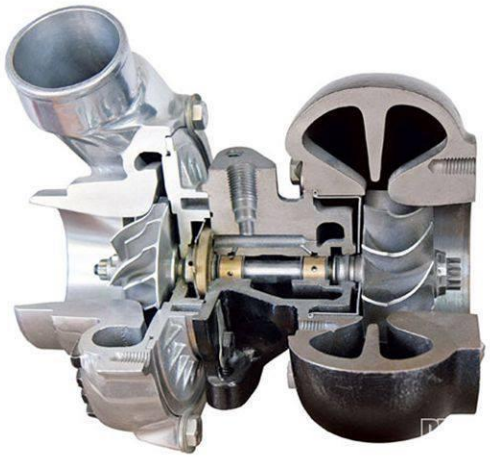




HYBRID AI IN A SINGLE EQUATION

Physics–Based Model

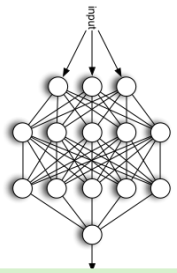




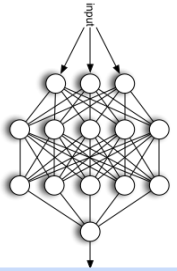
Physical asset

Simulation

Data collection



MACHINE LEARNING TECHNIQUES





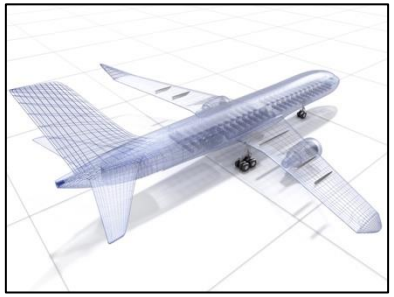
MOR

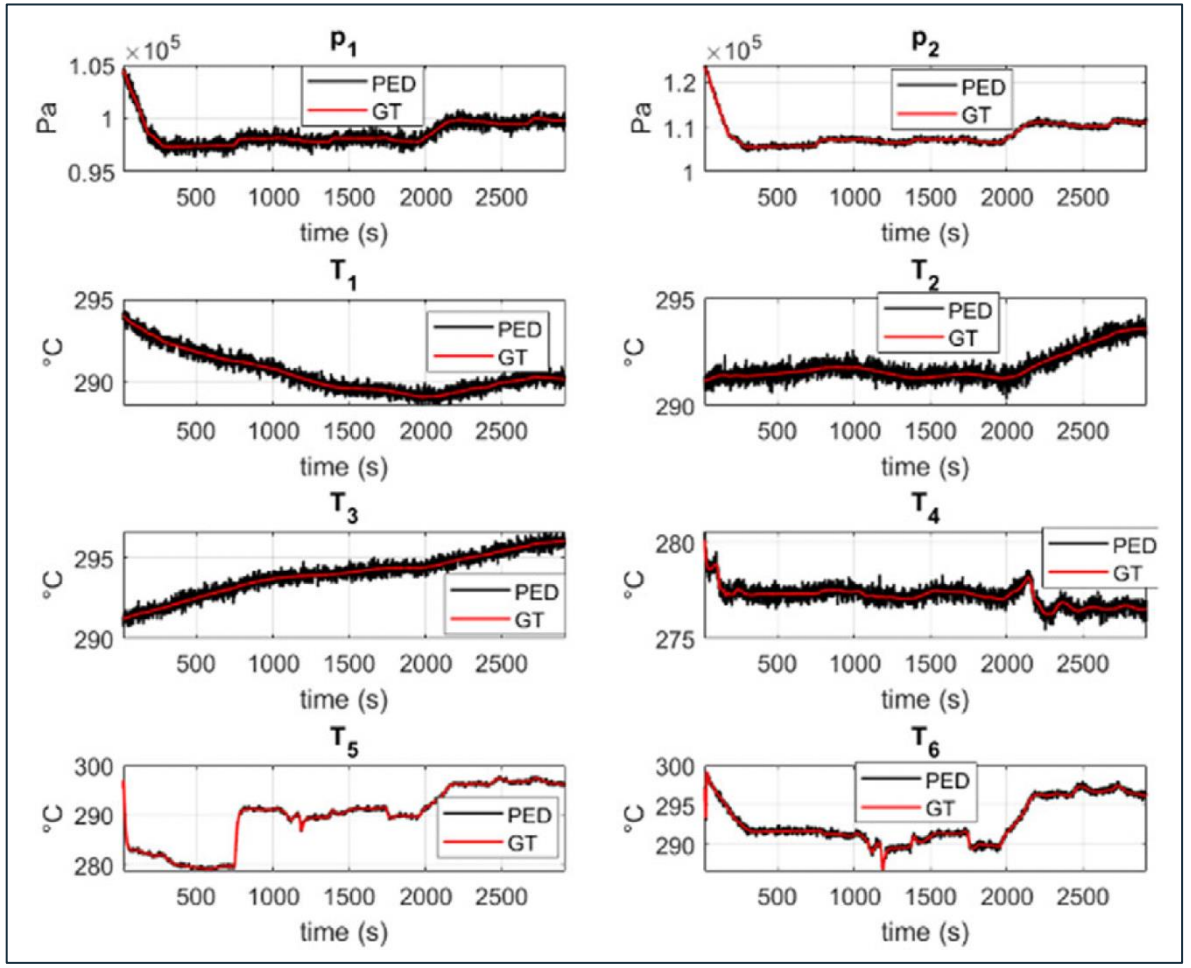
Ignorance model

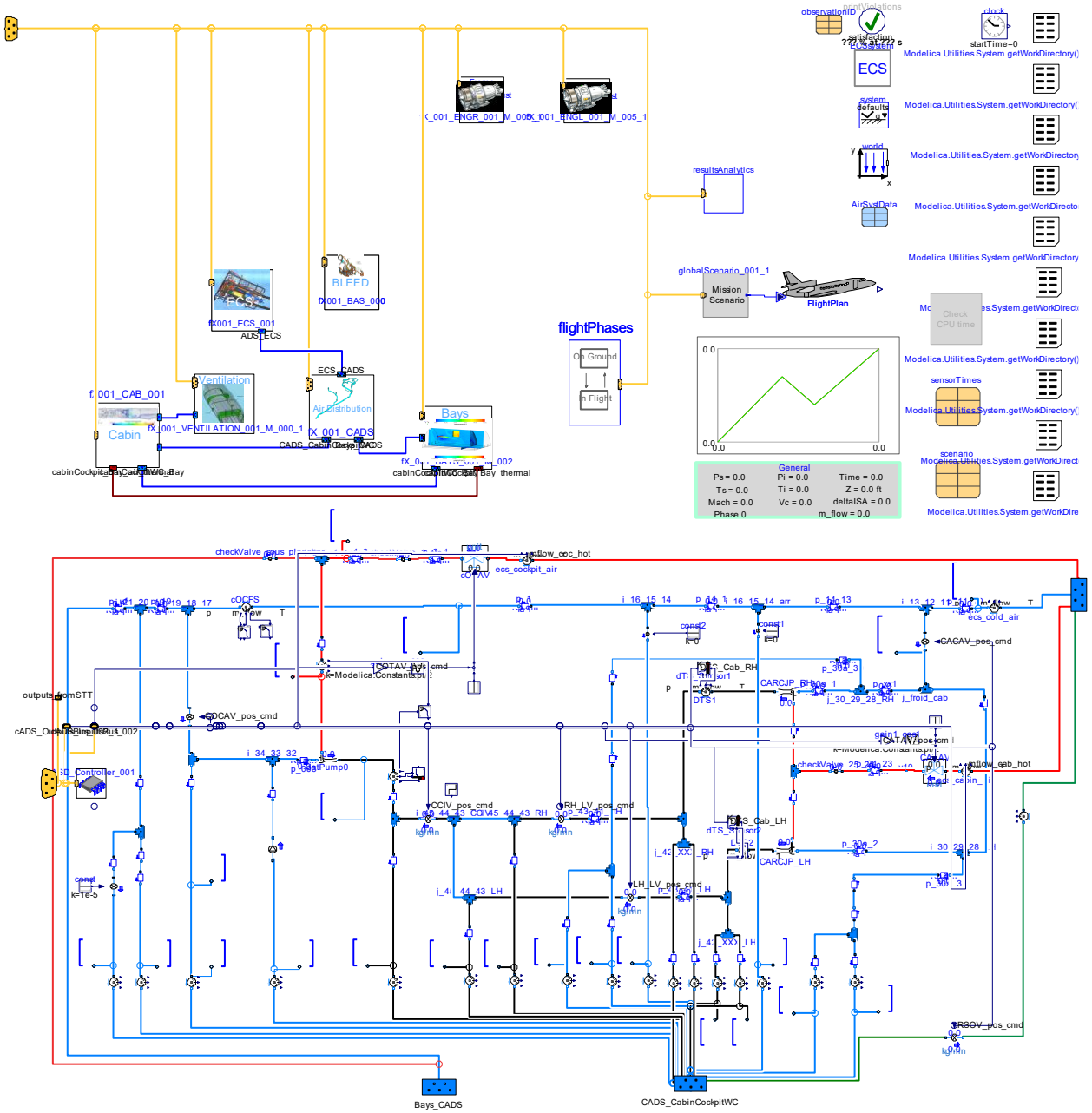
$$\frac{dx}{dt} = f^{\texttt{physics-based}}(x, \mathbf{z}) + f^{\texttt{data-driven}}(x, \mathbf{z})$$

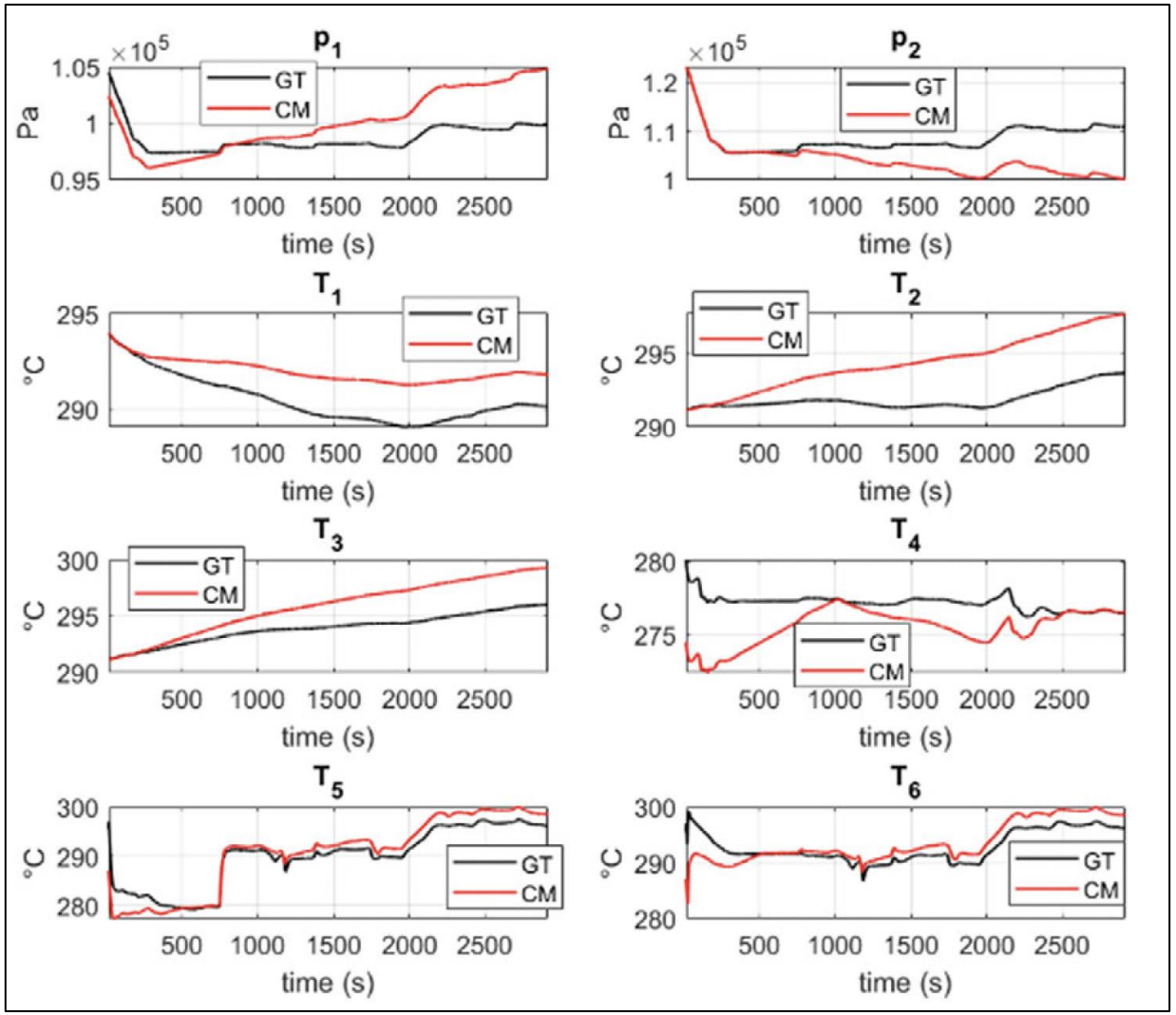
SYSTEM (larger) HYBRID (safer) TWIN

DASSAULT AVIATION

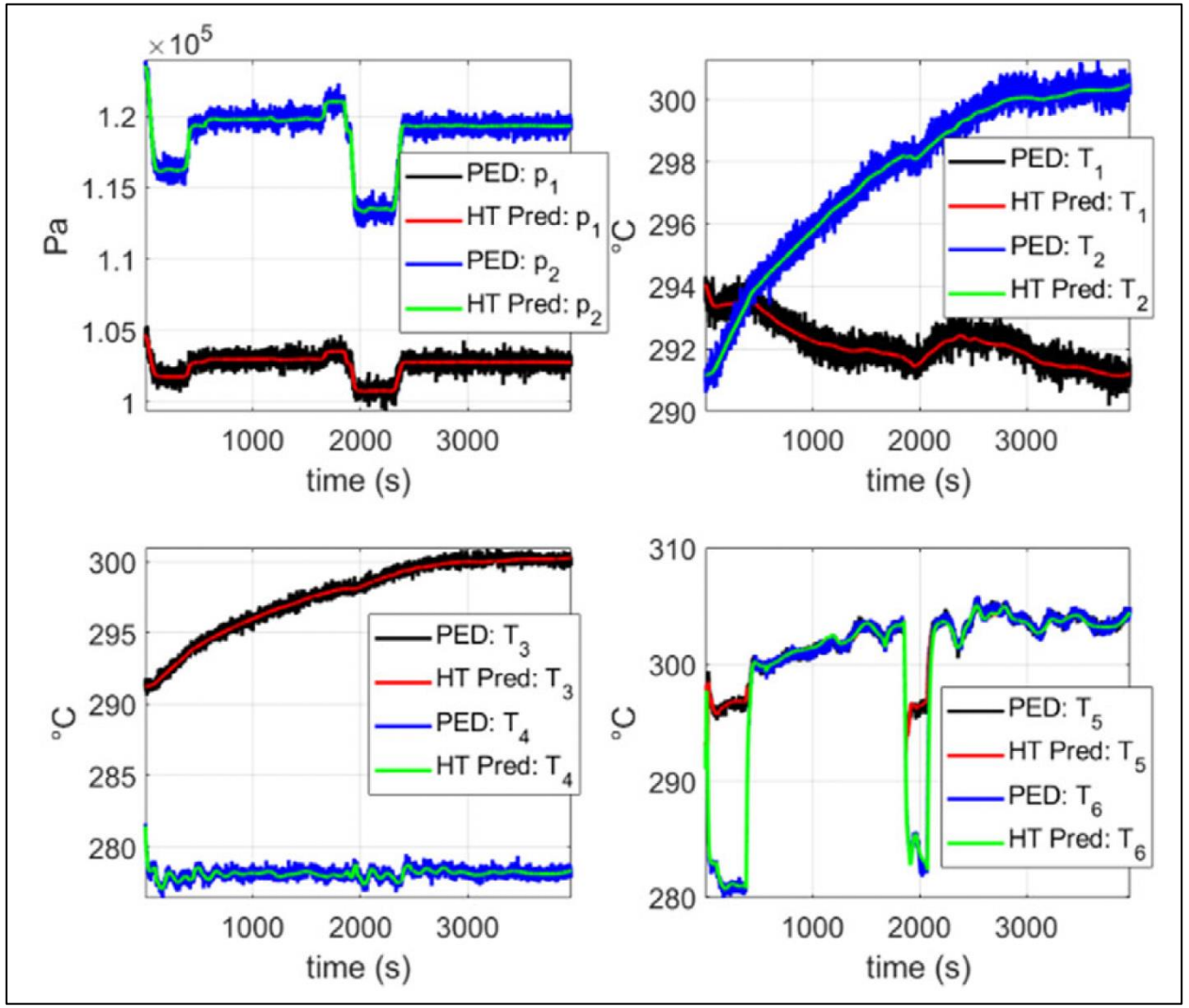




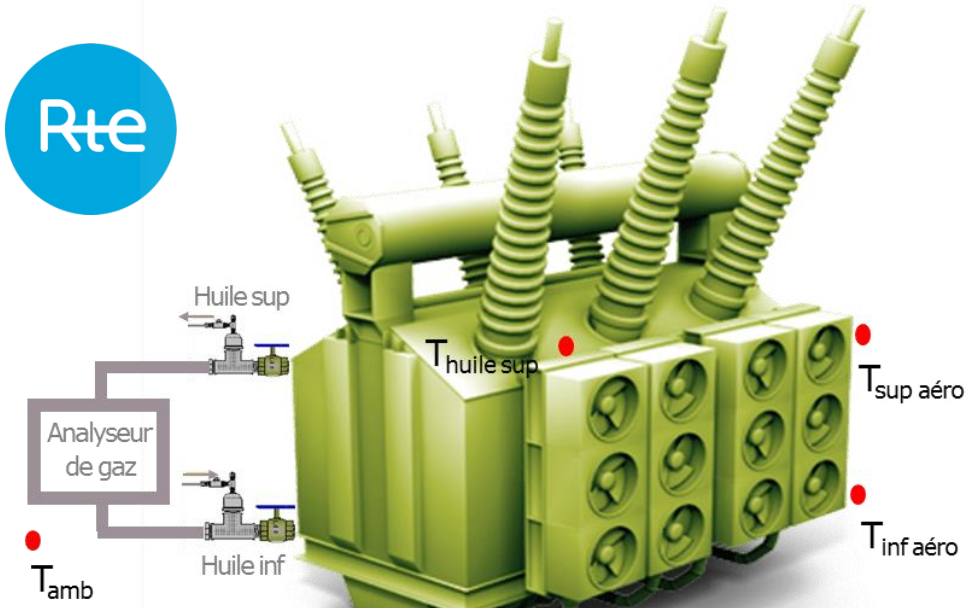


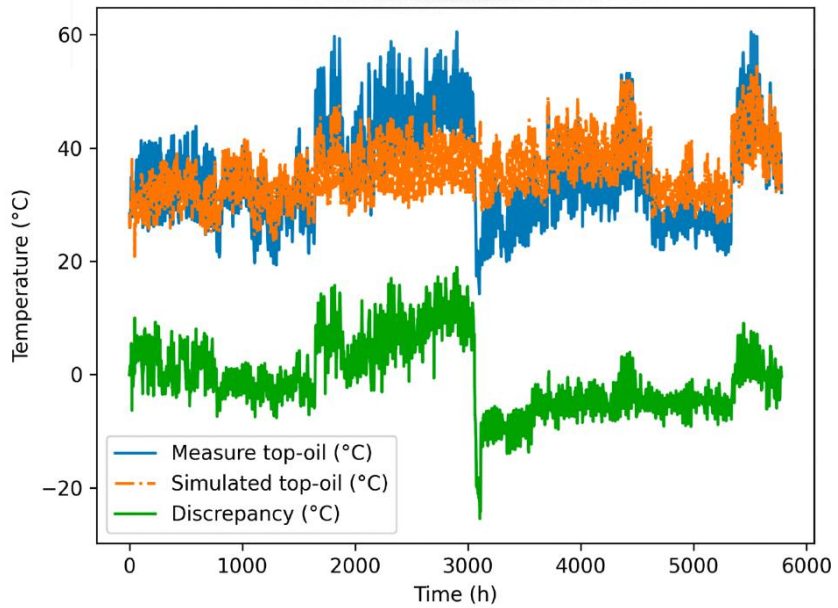


$$\frac{dx}{dt} = f^{\texttt{physics-based}}(x, \mathbf{z}) + f^{\texttt{data-driven}}(x, \mathbf{z})$$



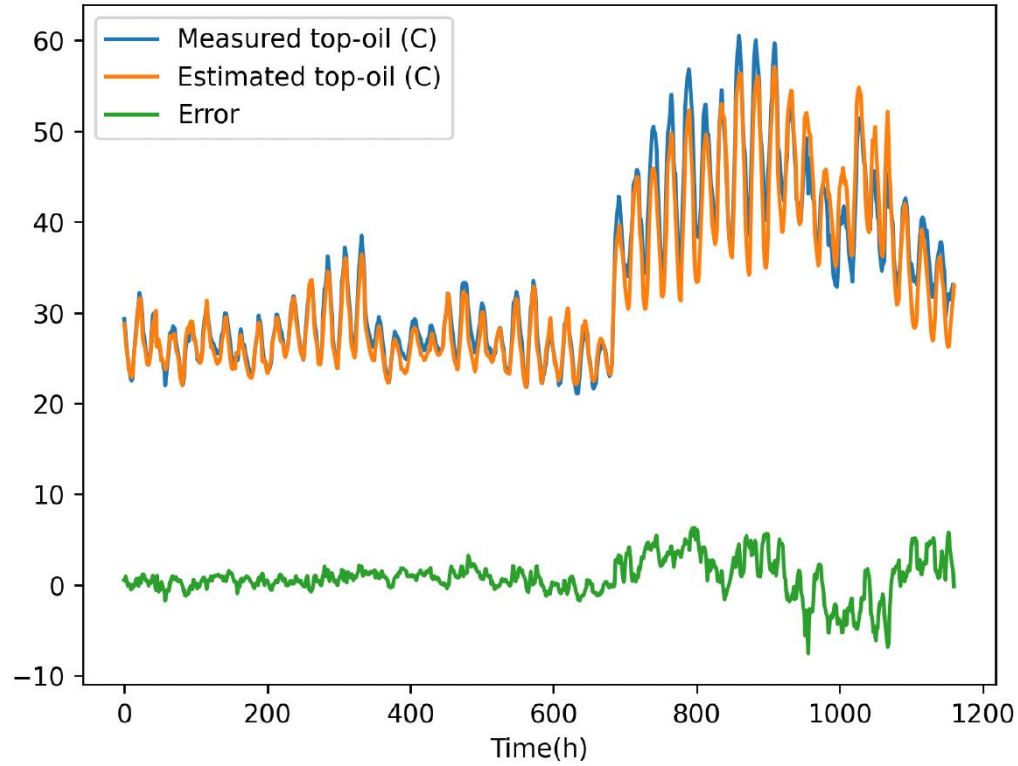
TRANFORMER HYBRID TWIN INSTANCE



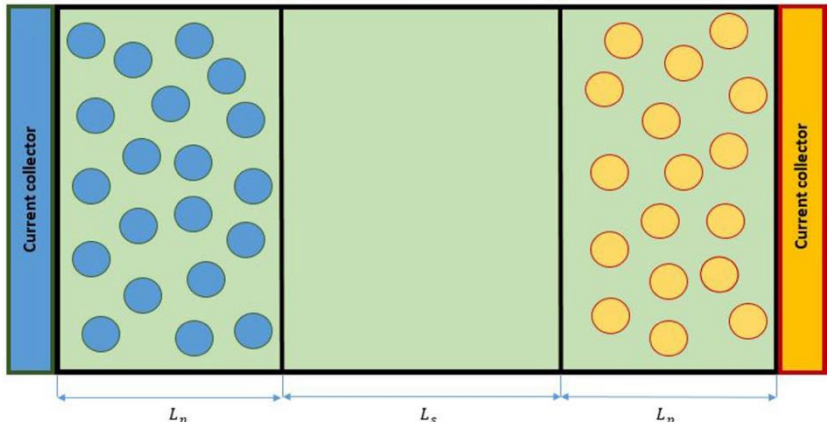


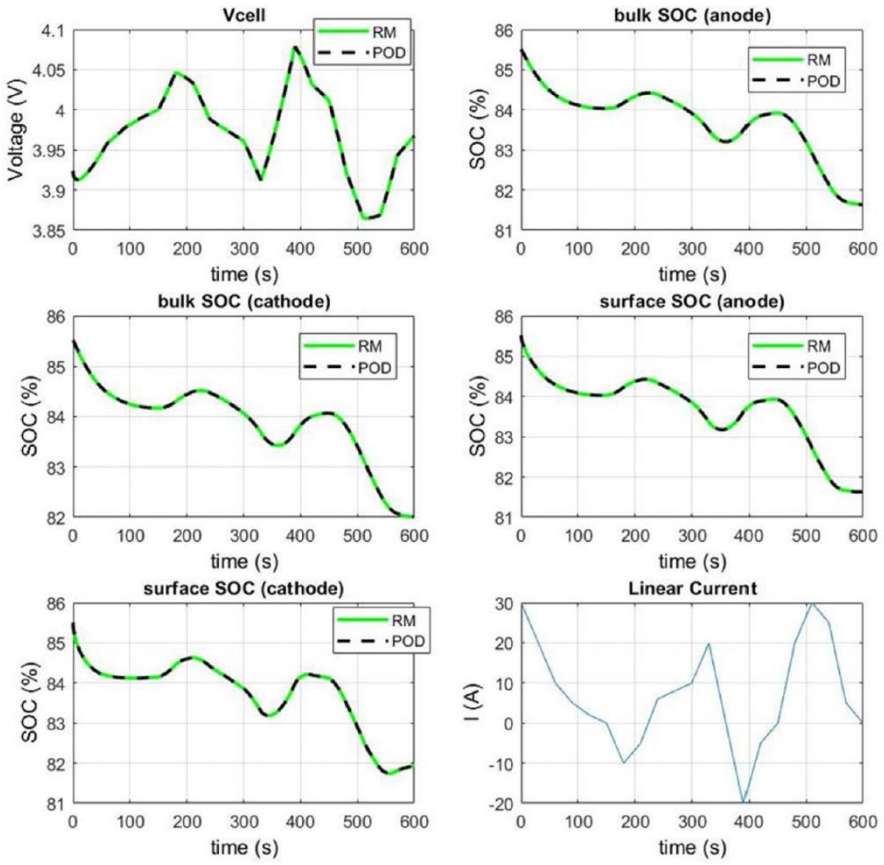
$$\frac{dx}{dt} = f^{\texttt{physics-based}}(x, \mathbf{z}) + f^{\texttt{data-driven}}(x, \mathbf{z})$$

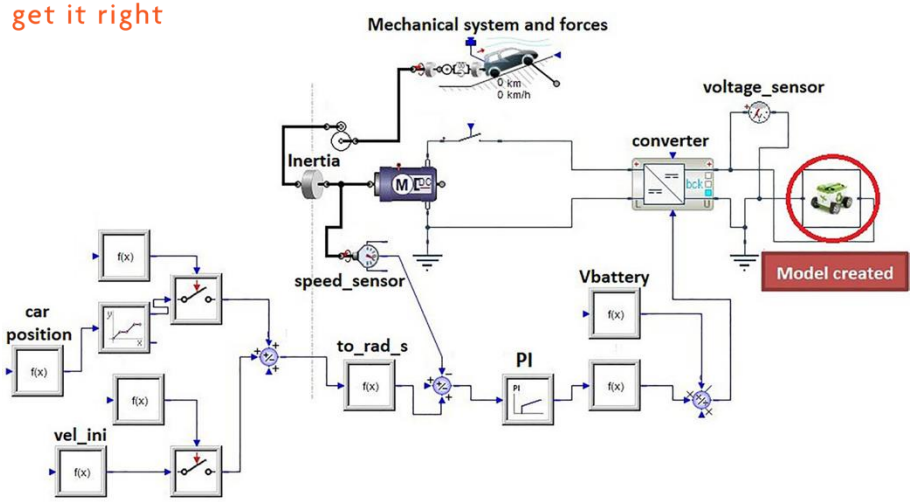
Testing+integration: HT Oil temperature estimation for a RTE transformer

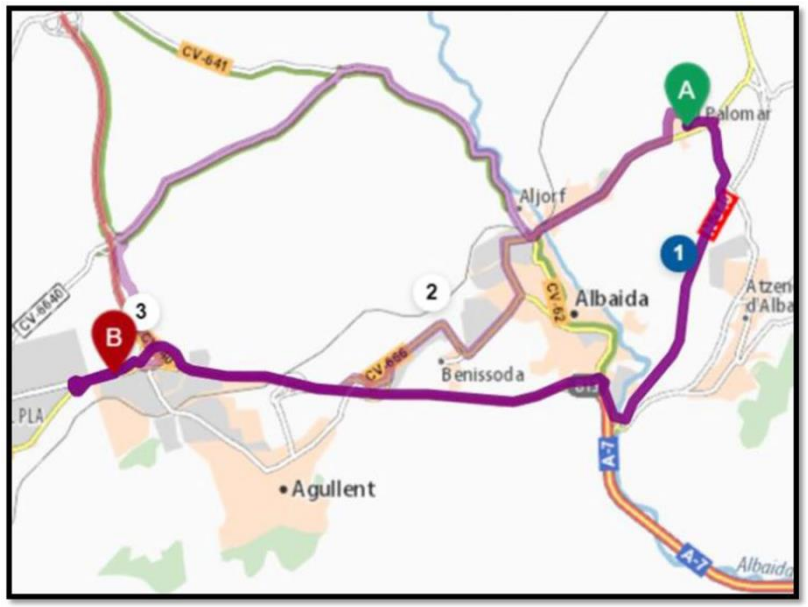


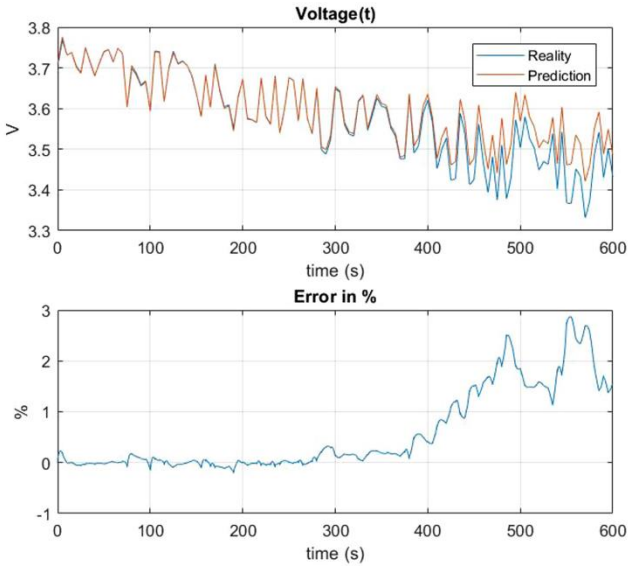
BATTERY HYBRID TWIN INSTANCE











HYBRID

$$\frac{dx}{dt} = f^{\texttt{physics-based}}(x, \mathbf{z}) + f^{\texttt{data-driven}}(x, \mathbf{z})$$

